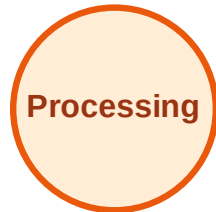


# BFS Traversal

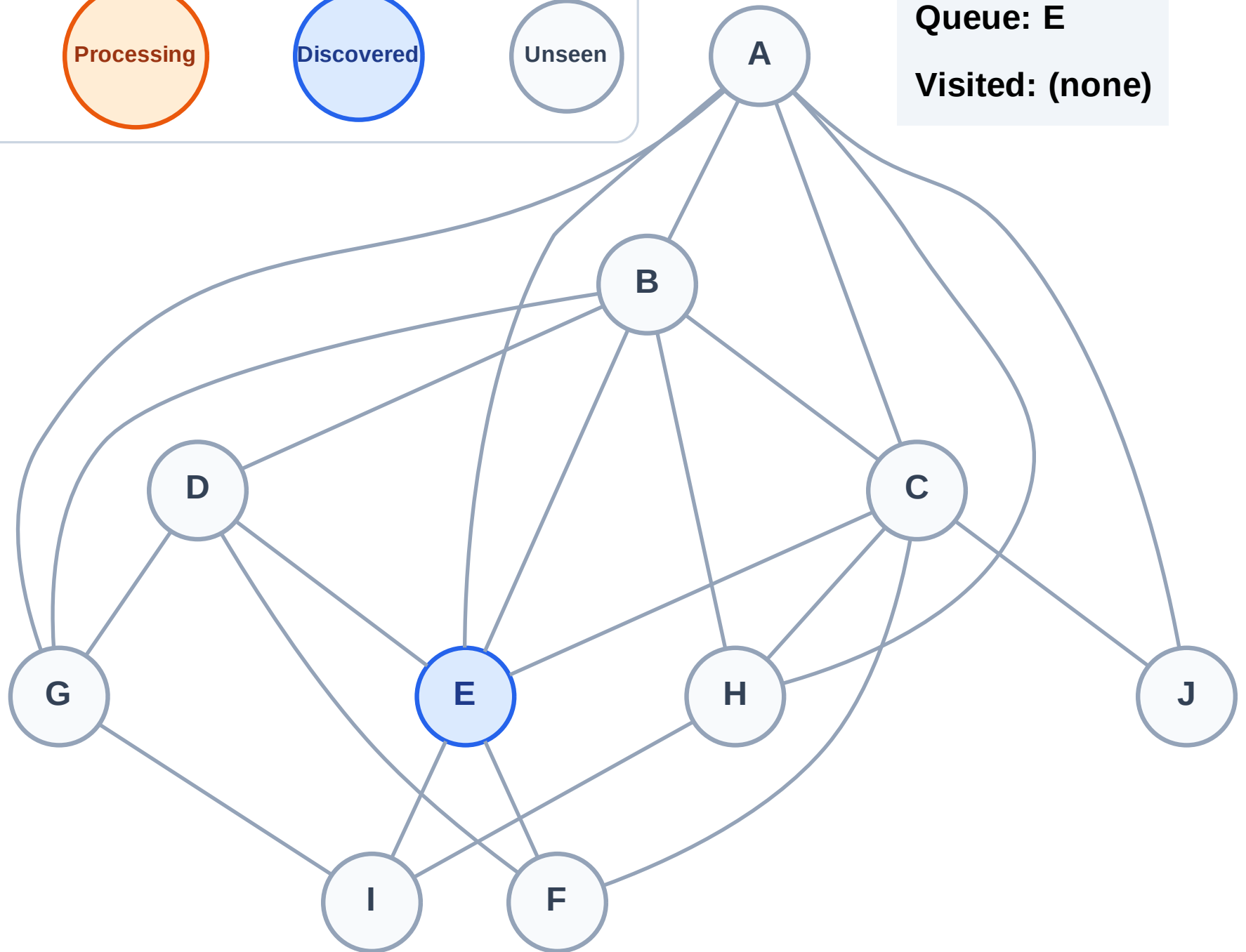
Step 1: Start at E

## Legend



Queue: E

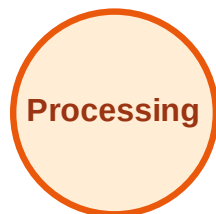
Visited: (none)



# BFS Traversal

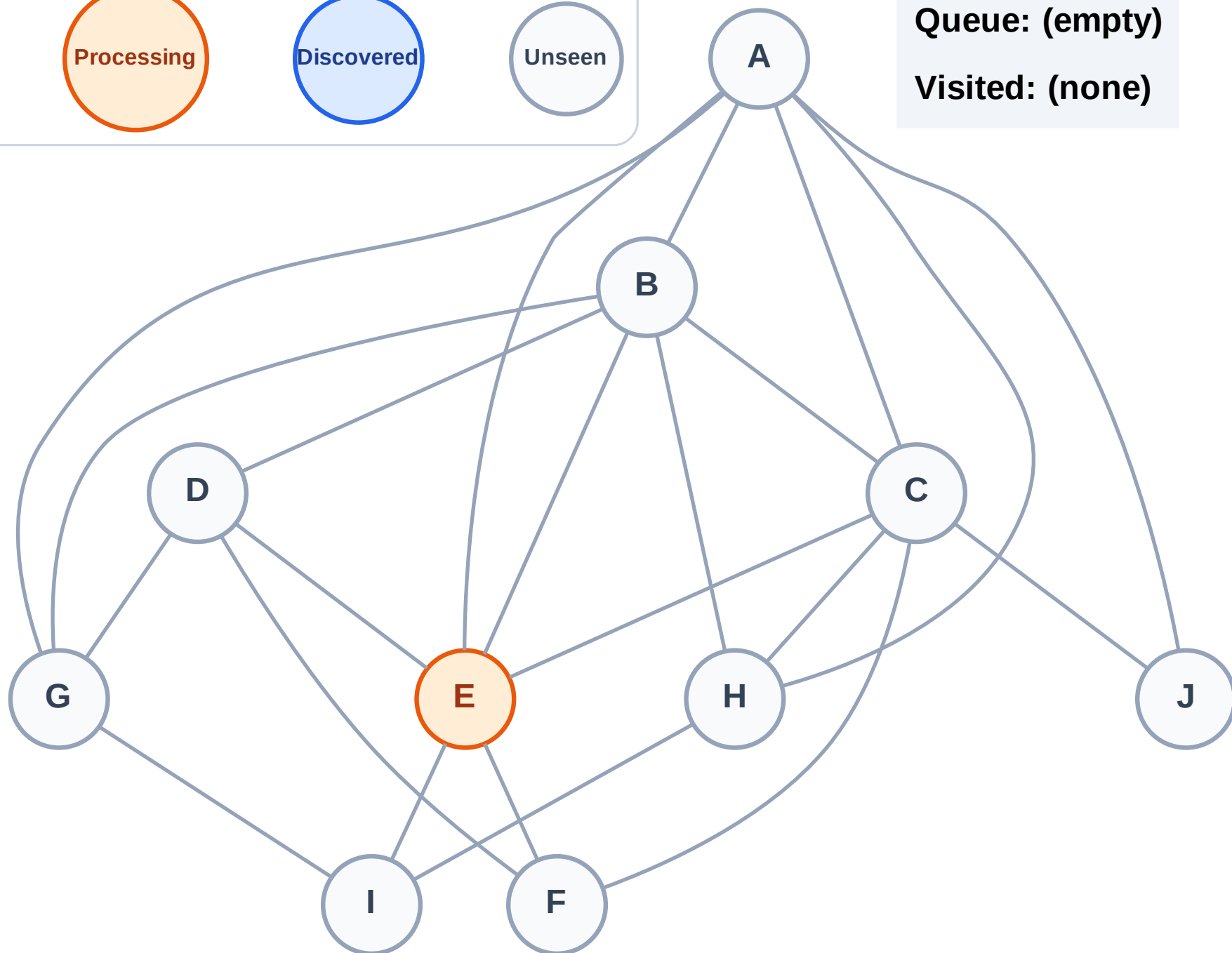
Step 2: Process node E

## Legend



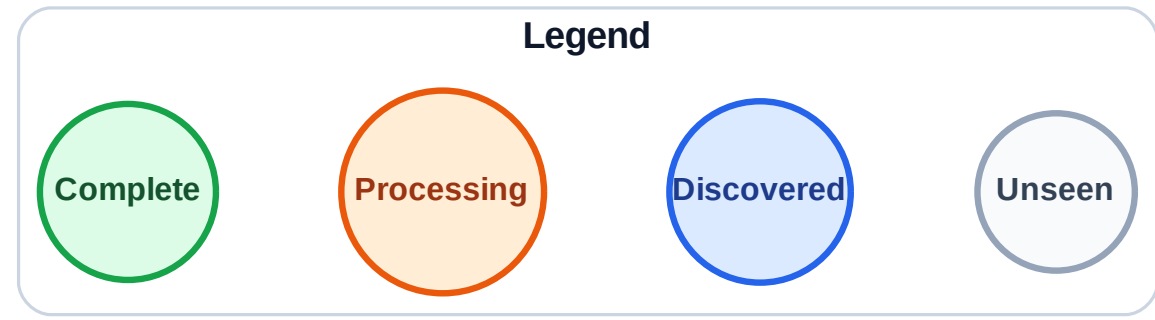
Queue: (empty)

Visited: (none)

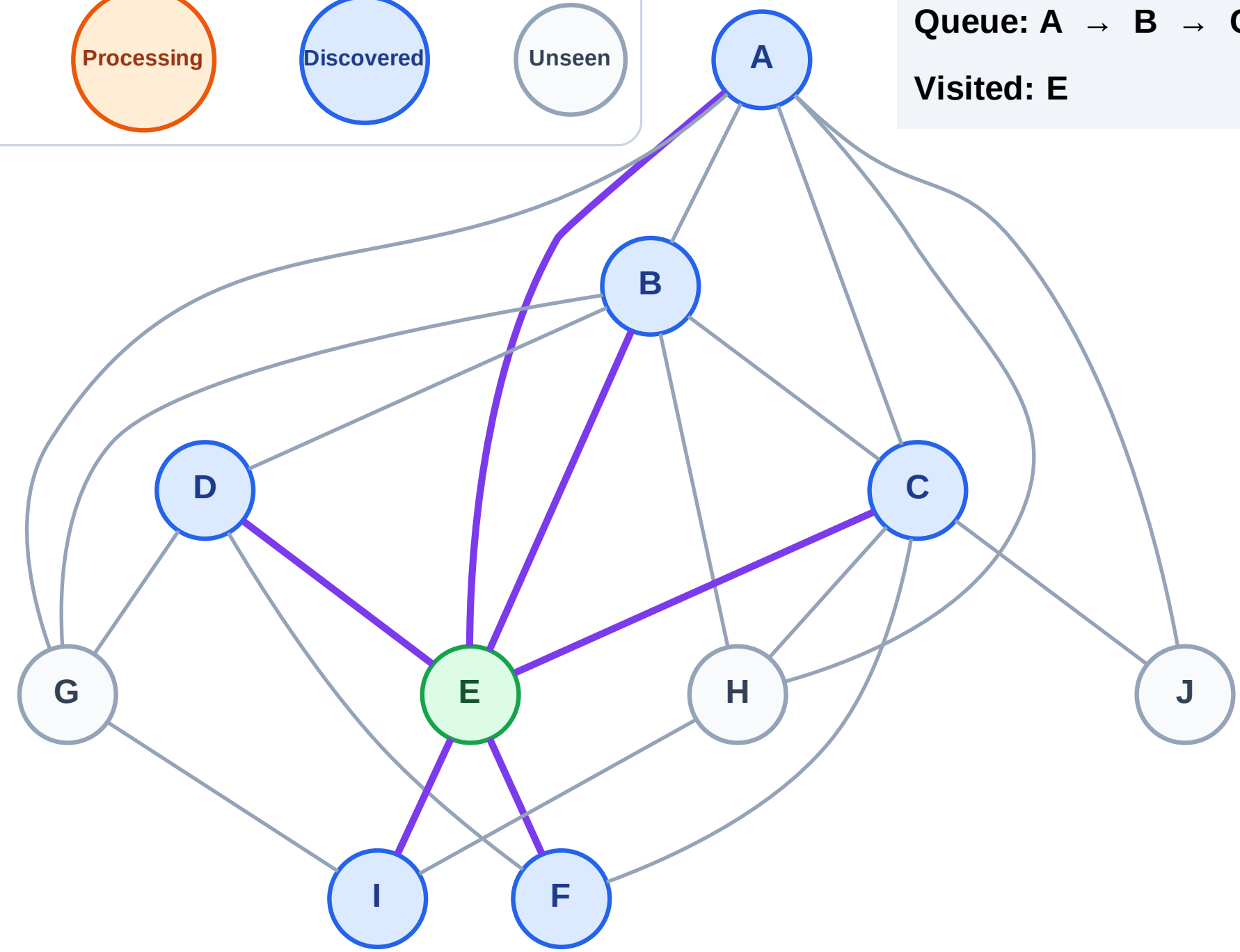


# BFS Traversal

Step 3: Discover A, B, C, D, F, I from E



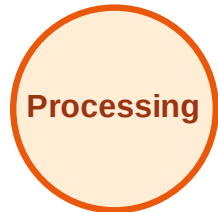
**Queue:** A → B → C → D → F → I  
**Visited:** E



# BFS Traversal

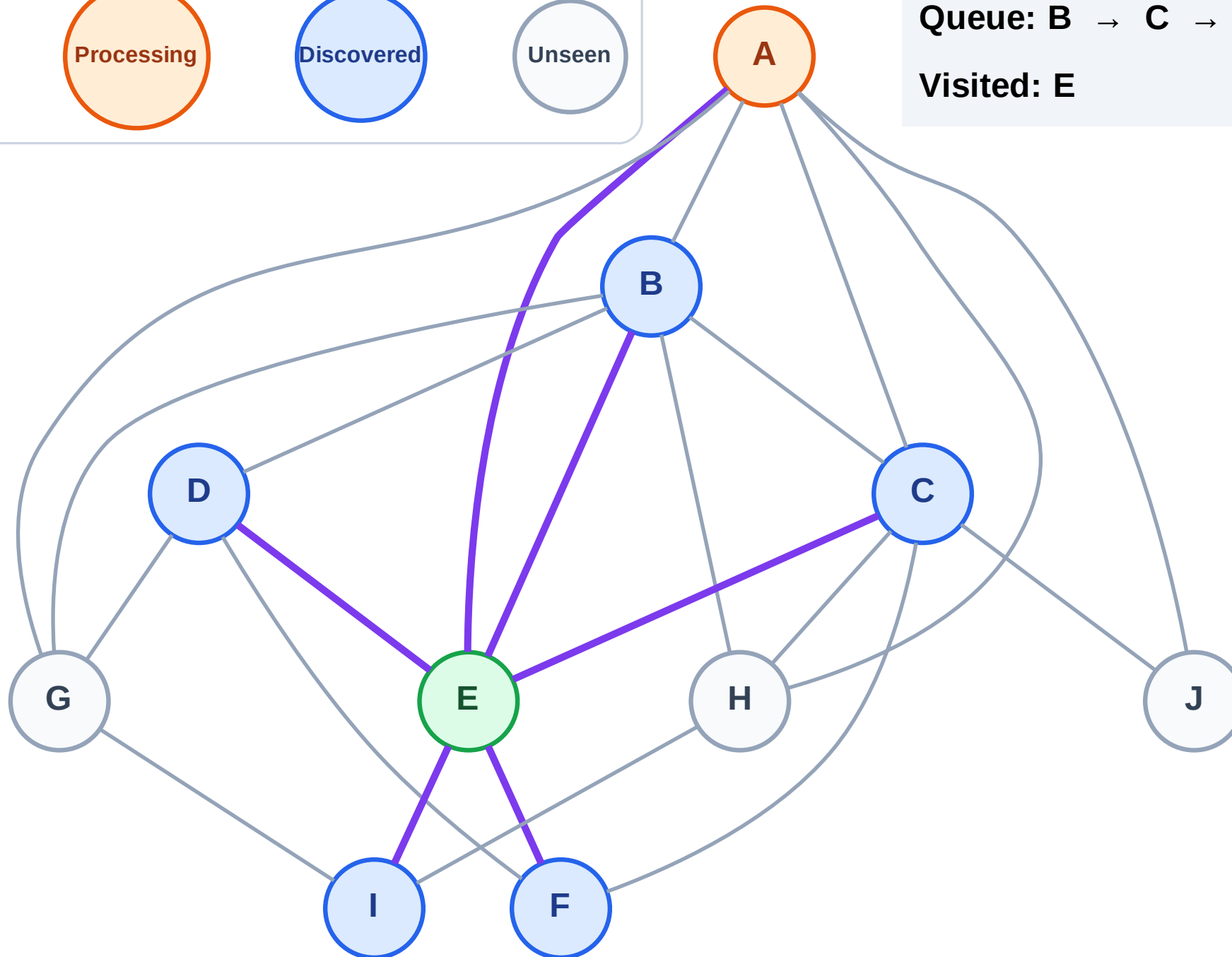
Step 4: Process node A

## Legend



Queue: B → C → D → F → I

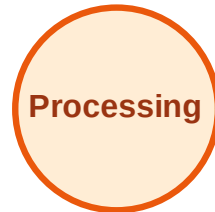
Visited: E



# BFS Traversal

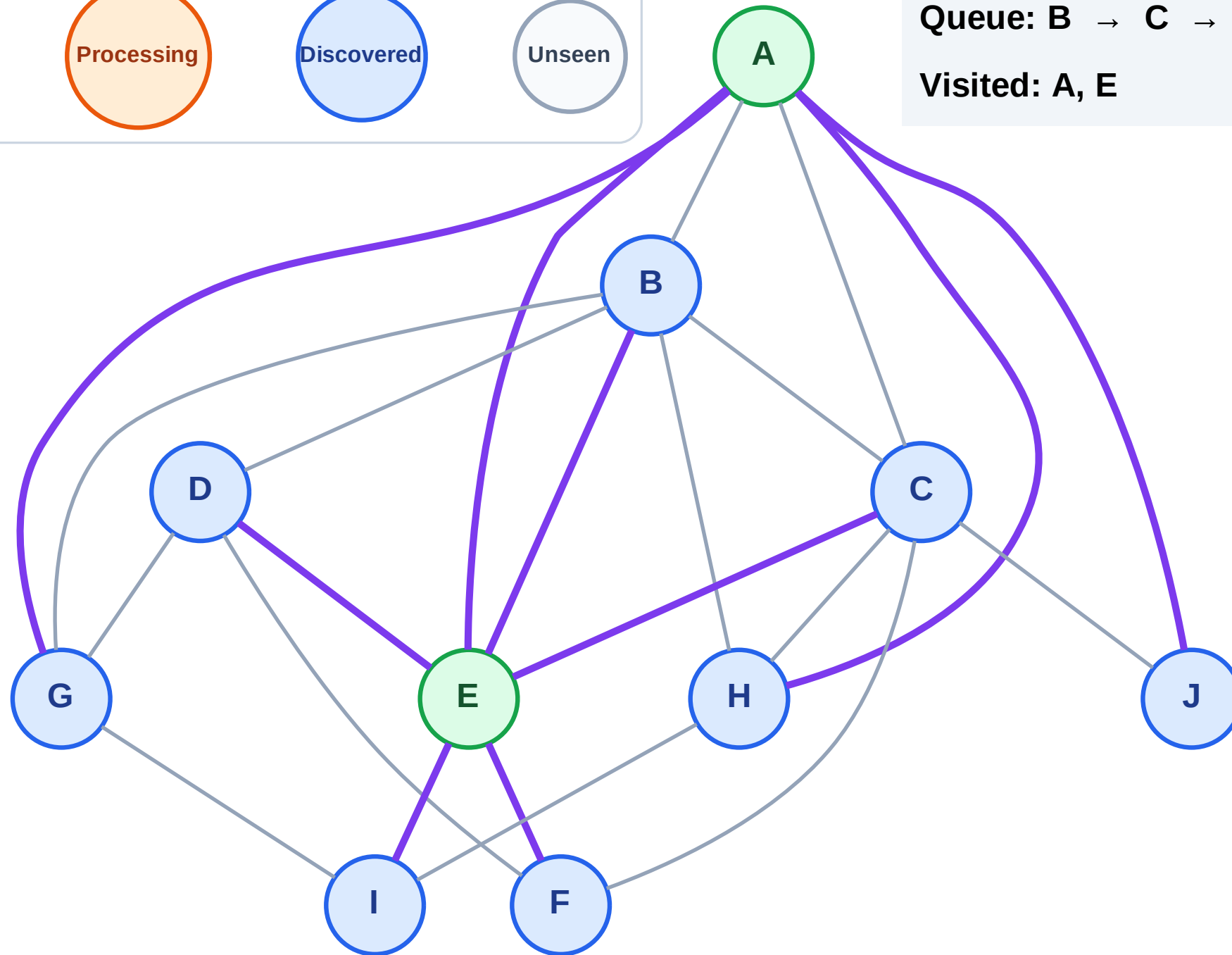
Step 5: Discover G, H, J from A

## Legend



Queue: B → C → D → F → I → G → H → J

Visited: A, E



# BFS Traversal

Step 6: Process node B

Legend

Complete

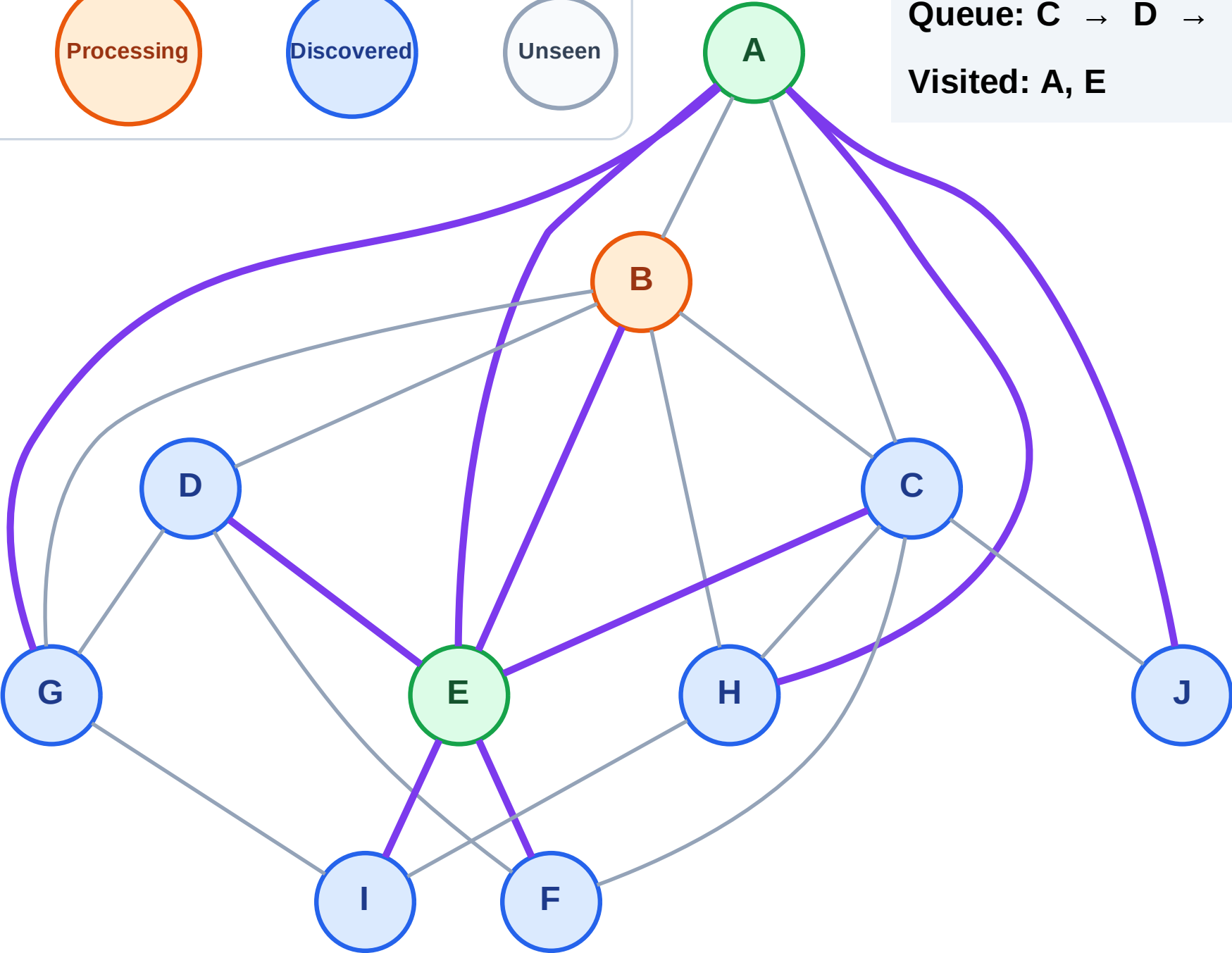
Processing

Discovered

Unseen

Queue: C → D → F → I → G → H → J

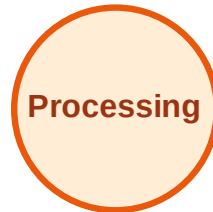
Visited: A, E



# BFS Traversal

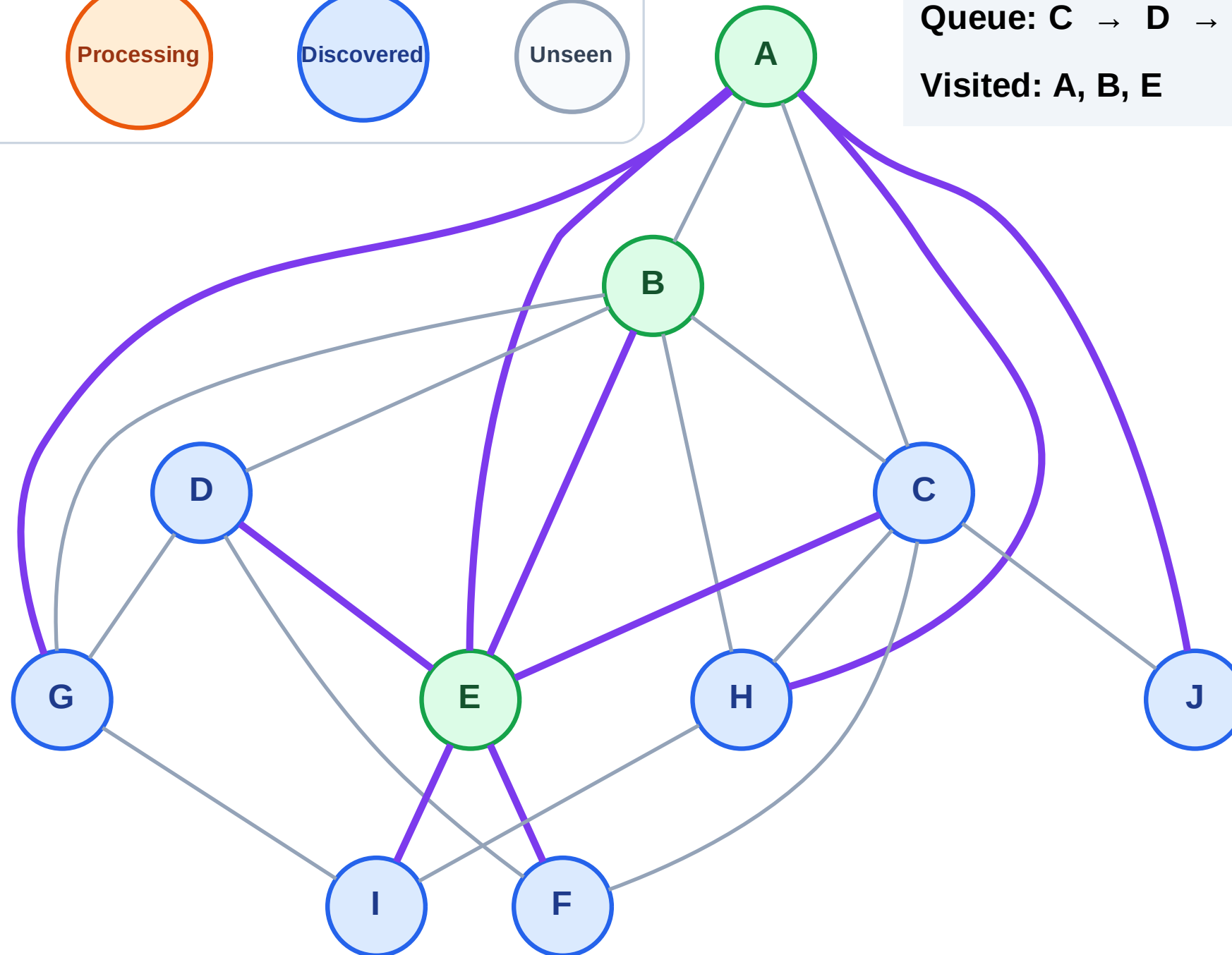
Step 7: No new neighbors from B

## Legend



Queue: C → D → F → I → G → H → J

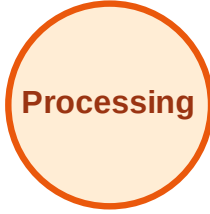
Visited: A, B, E



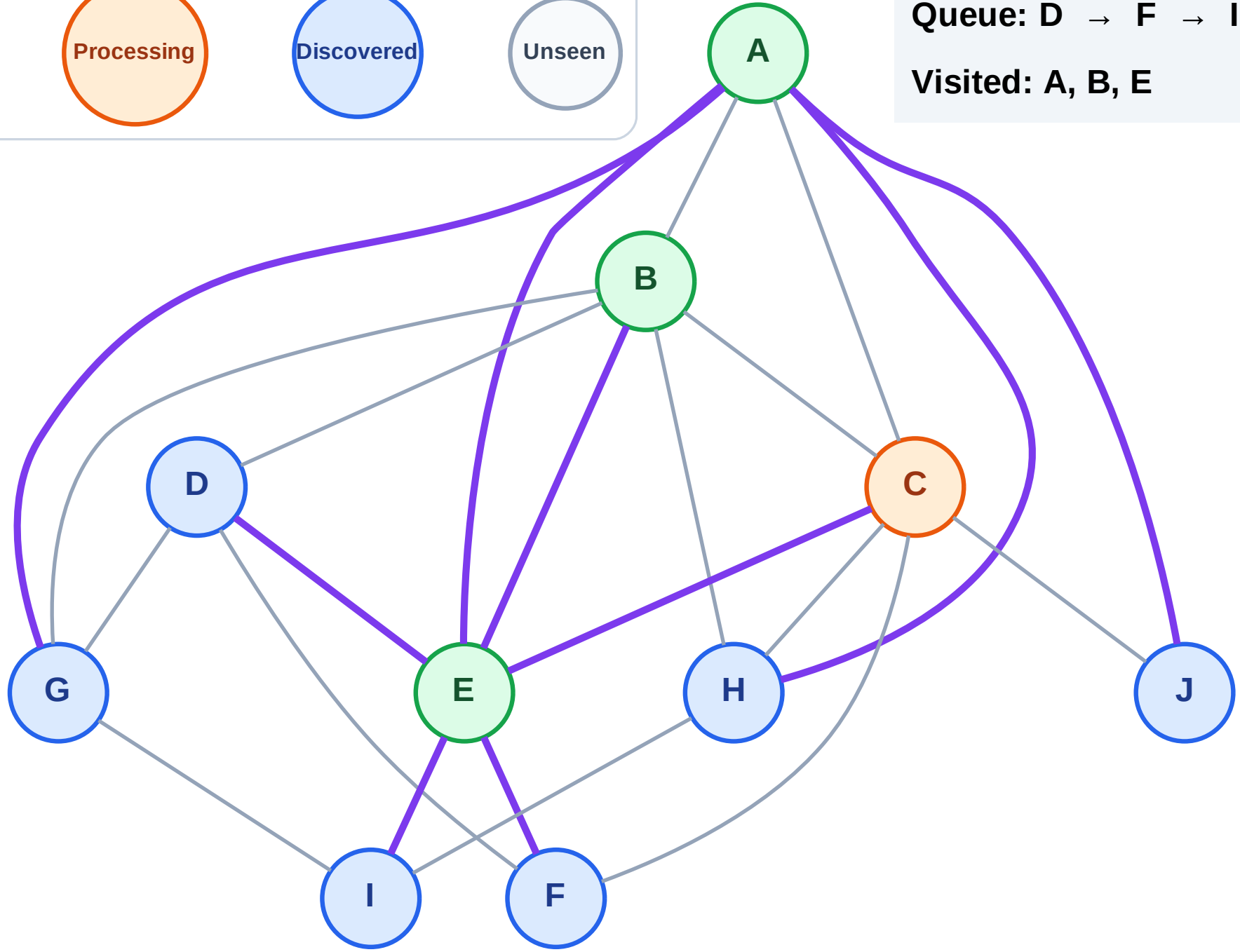
# BFS Traversal

Step 8: Process node C

## Legend



Queue: D → F → I → G → H → J  
Visited: A, B, E

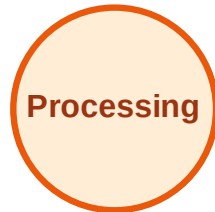




# BFS Traversal

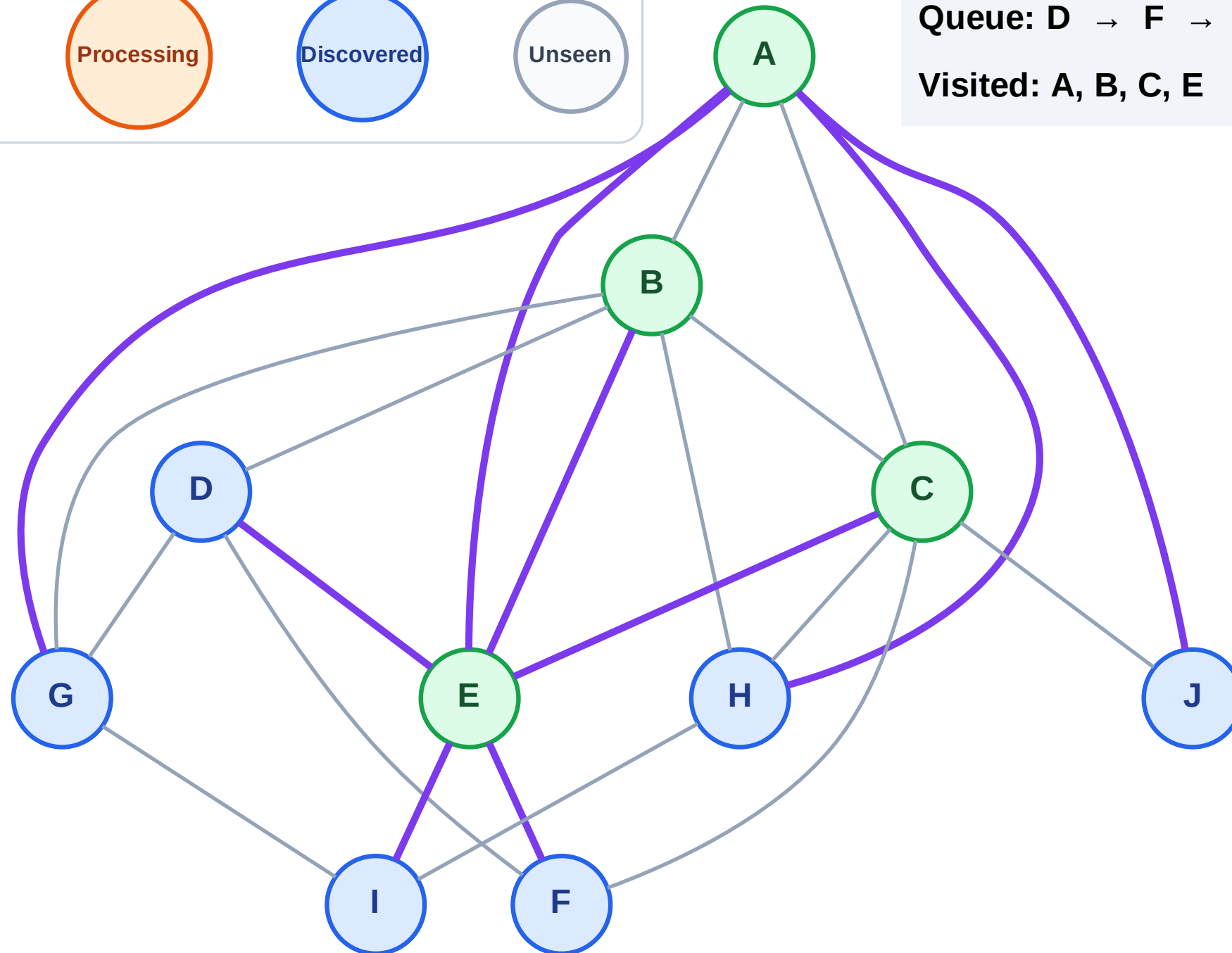
Step 9: No new neighbors from C

## Legend



Queue: D → F → I → G → H → J

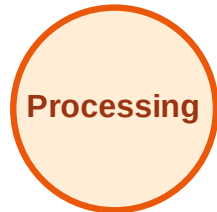
Visited: A, B, C, E



# BFS Traversal

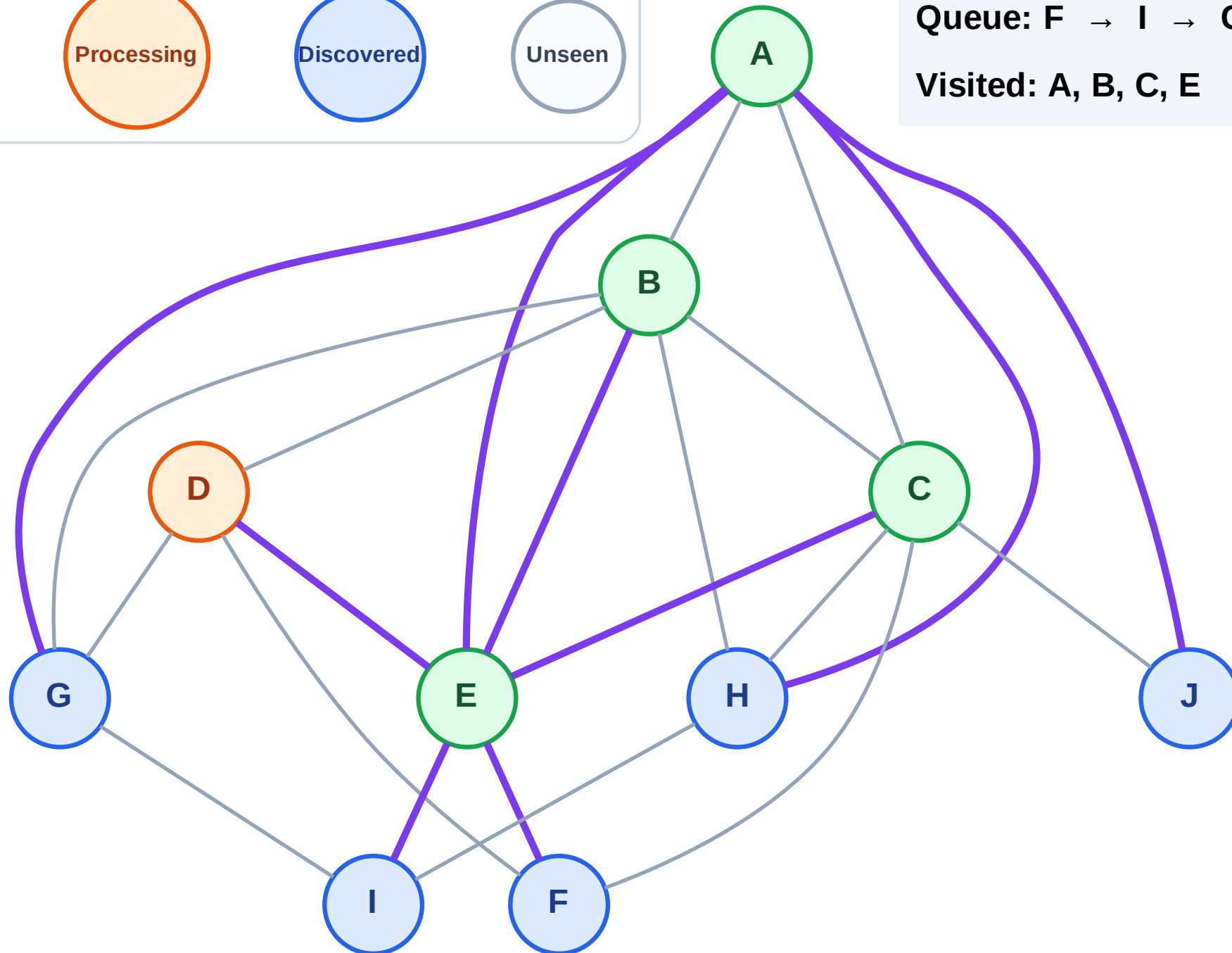
Step 10: Process node D

## Legend



Queue: F → I → G → H → J

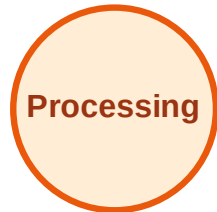
Visited: A, B, C, E



# BFS Traversal

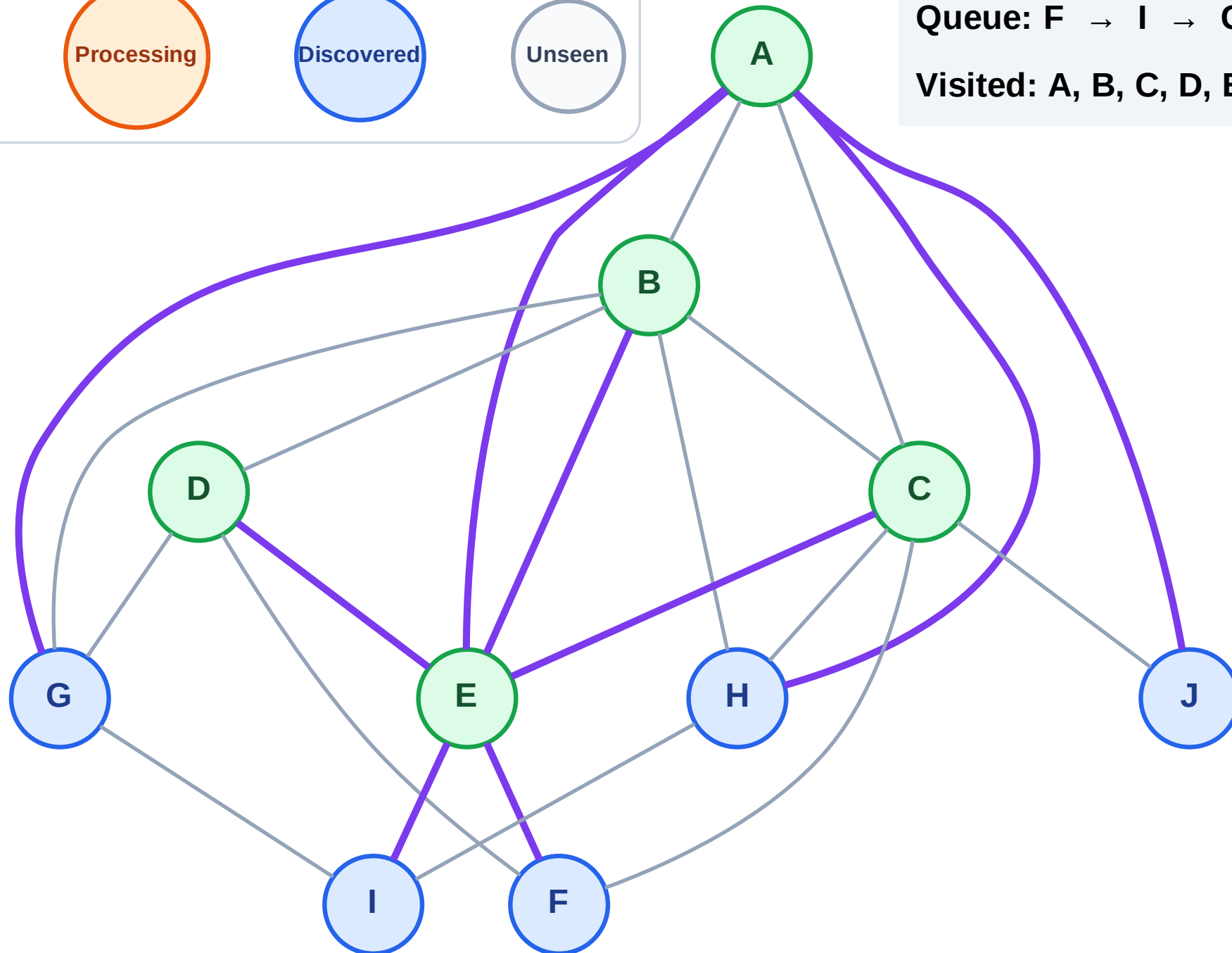
Step 11: No new neighbors from D

## Legend



Queue: F → I → G → H → J

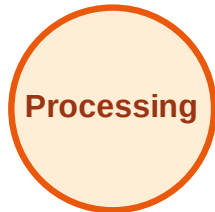
Visited: A, B, C, D, E



# BFS Traversal

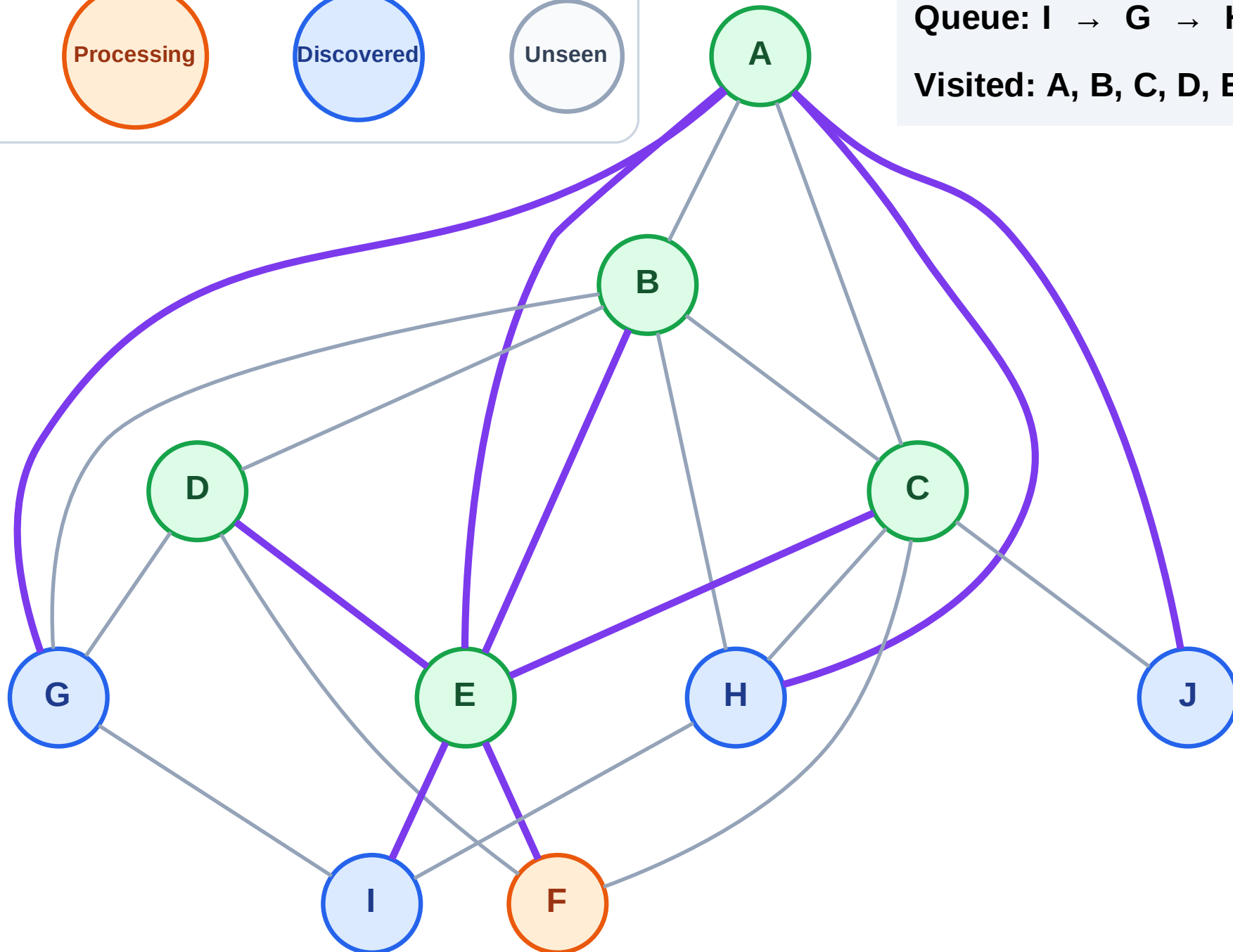
Step 12: Process node F

## Legend



Queue: I → G → H → J

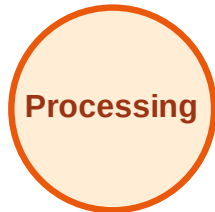
Visited: A, B, C, D, E



# BFS Traversal

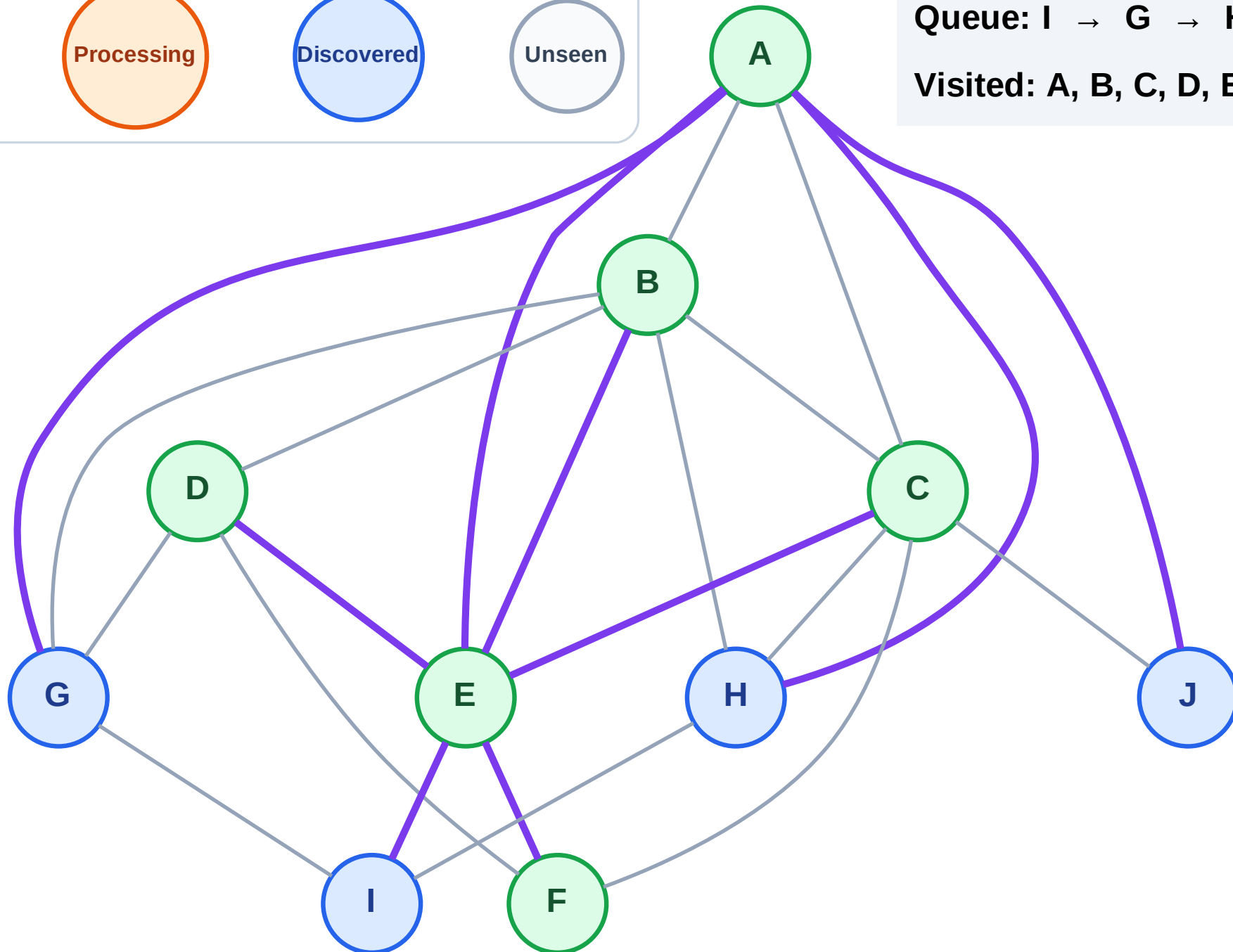
Step 13: No new neighbors from F

## Legend



Queue: I → G → H → J

Visited: A, B, C, D, E, F



# BFS Traversal

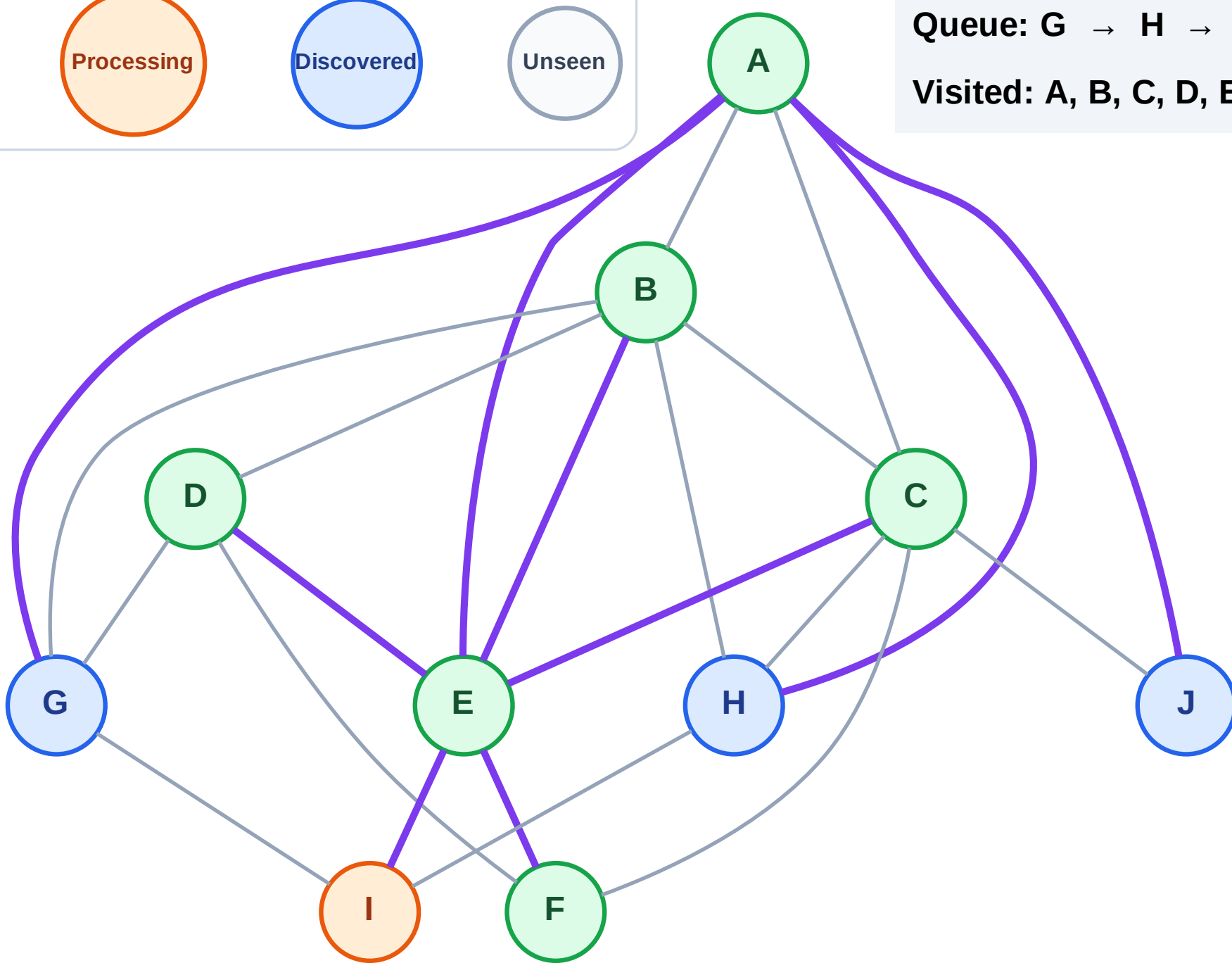
Step 14: Process node I

## Legend



Queue: G → H → J

Visited: A, B, C, D, E, F



# BFS Traversal

Step 15: No new neighbors from I

## Legend

Complete

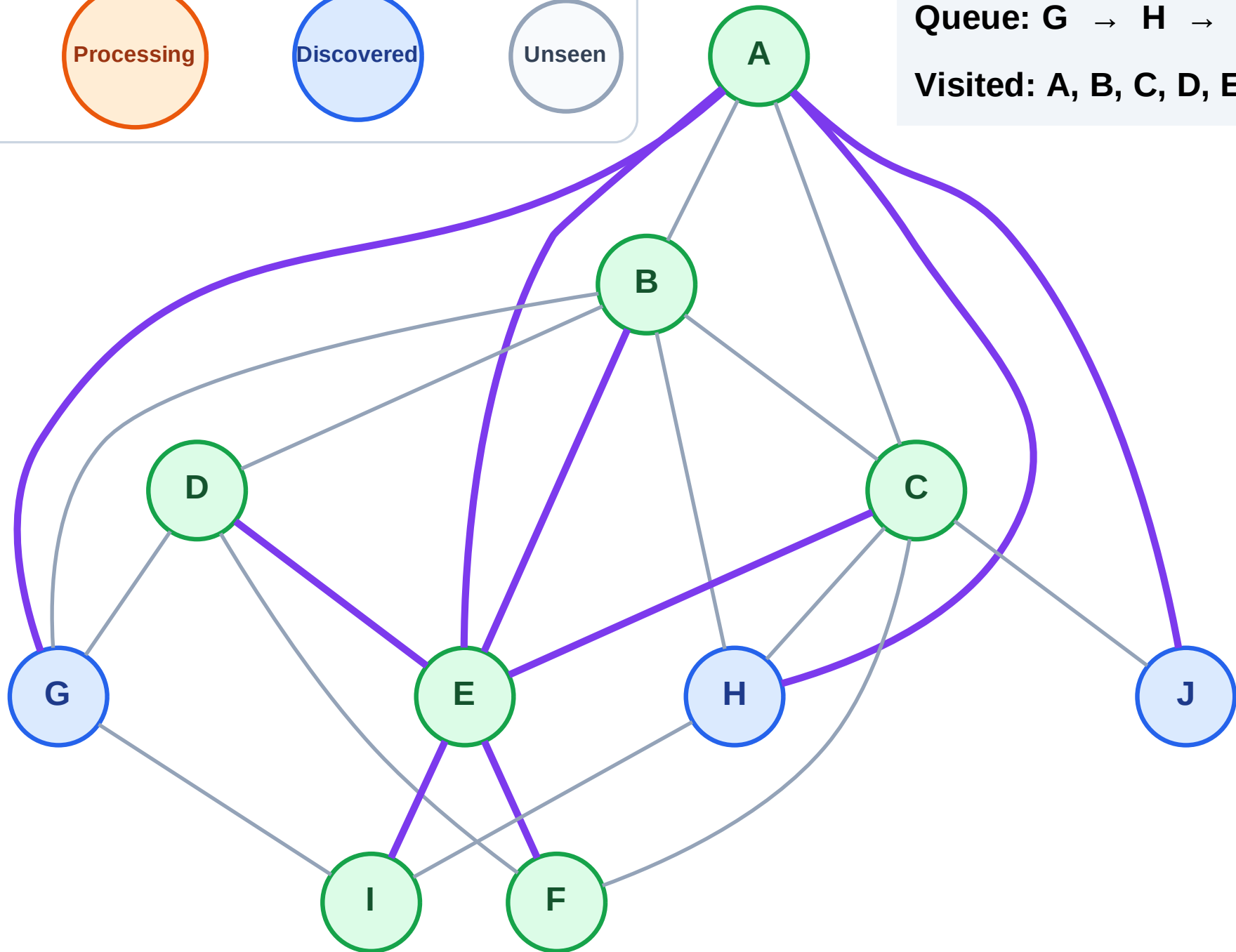
Processing

Discovered

Unseen

Queue: G → H → J

Visited: A, B, C, D, E, F, I



# BFS Traversal

Step 16: Process node G

## Legend

Complete

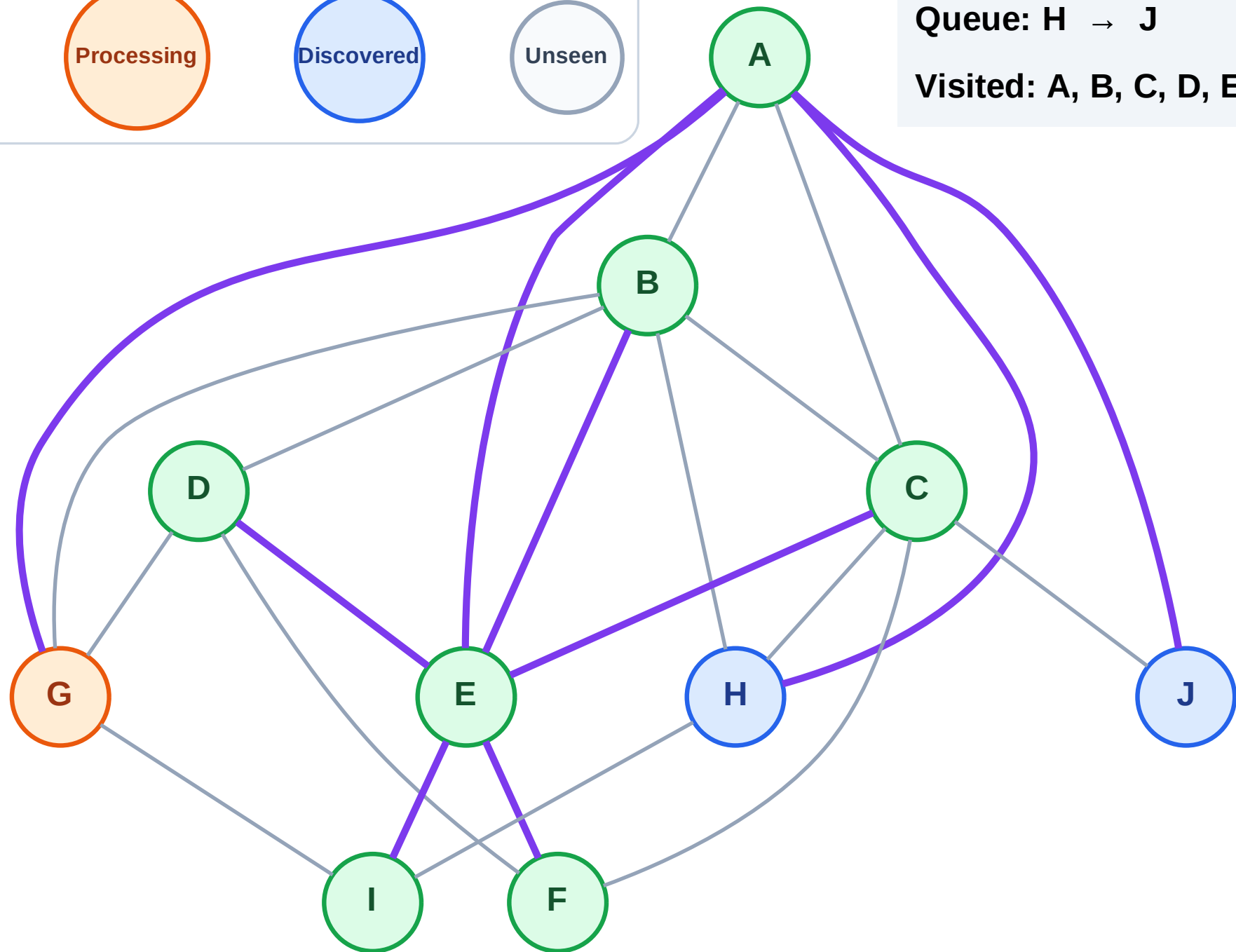
Processing

Discovered

Unseen

Queue: H → J

Visited: A, B, C, D, E, F, I





# BFS Traversal

Step 17: No new neighbors from G

## Legend

Complete

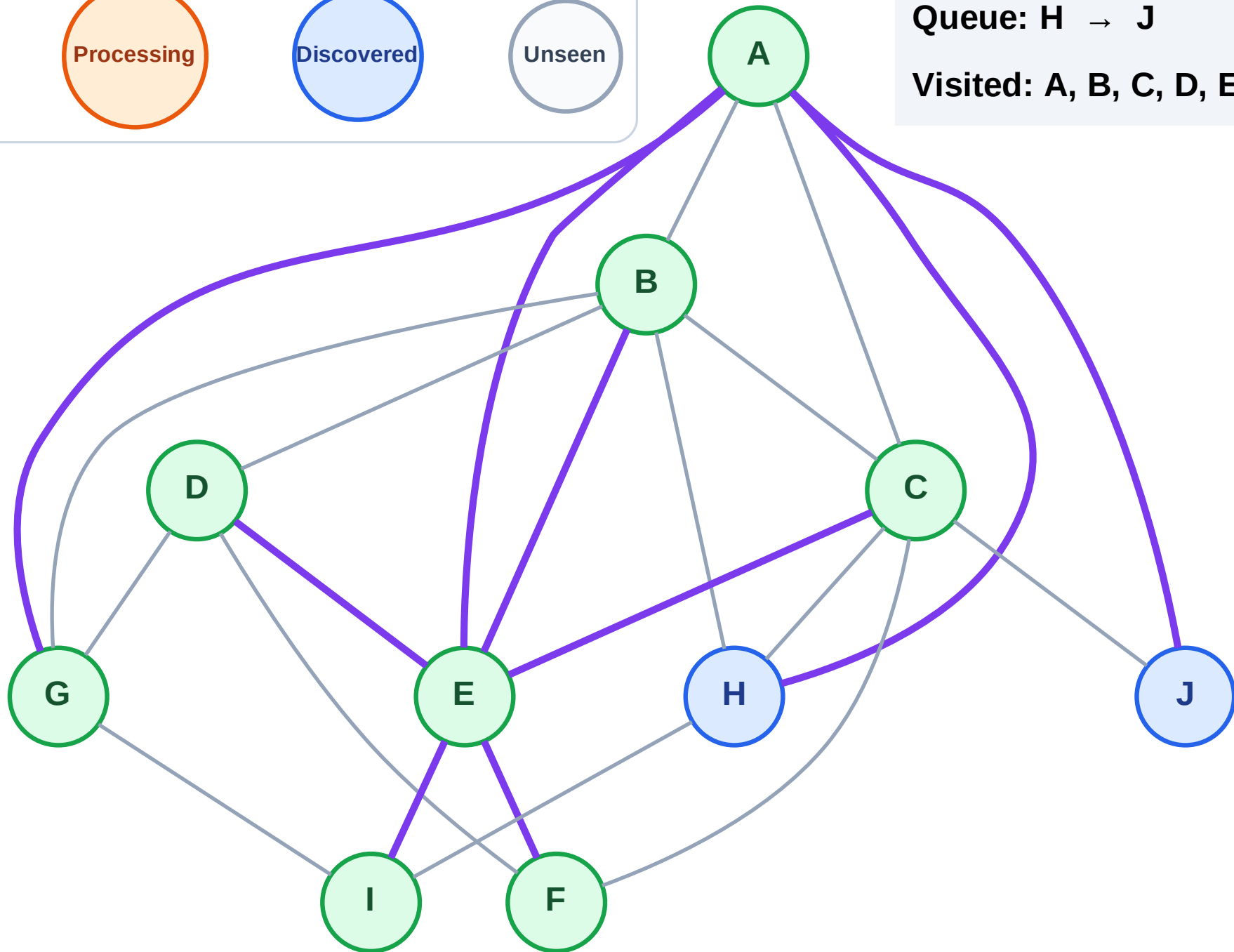
Processing

Discovered

Unseen

Queue: H → J

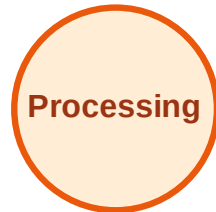
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

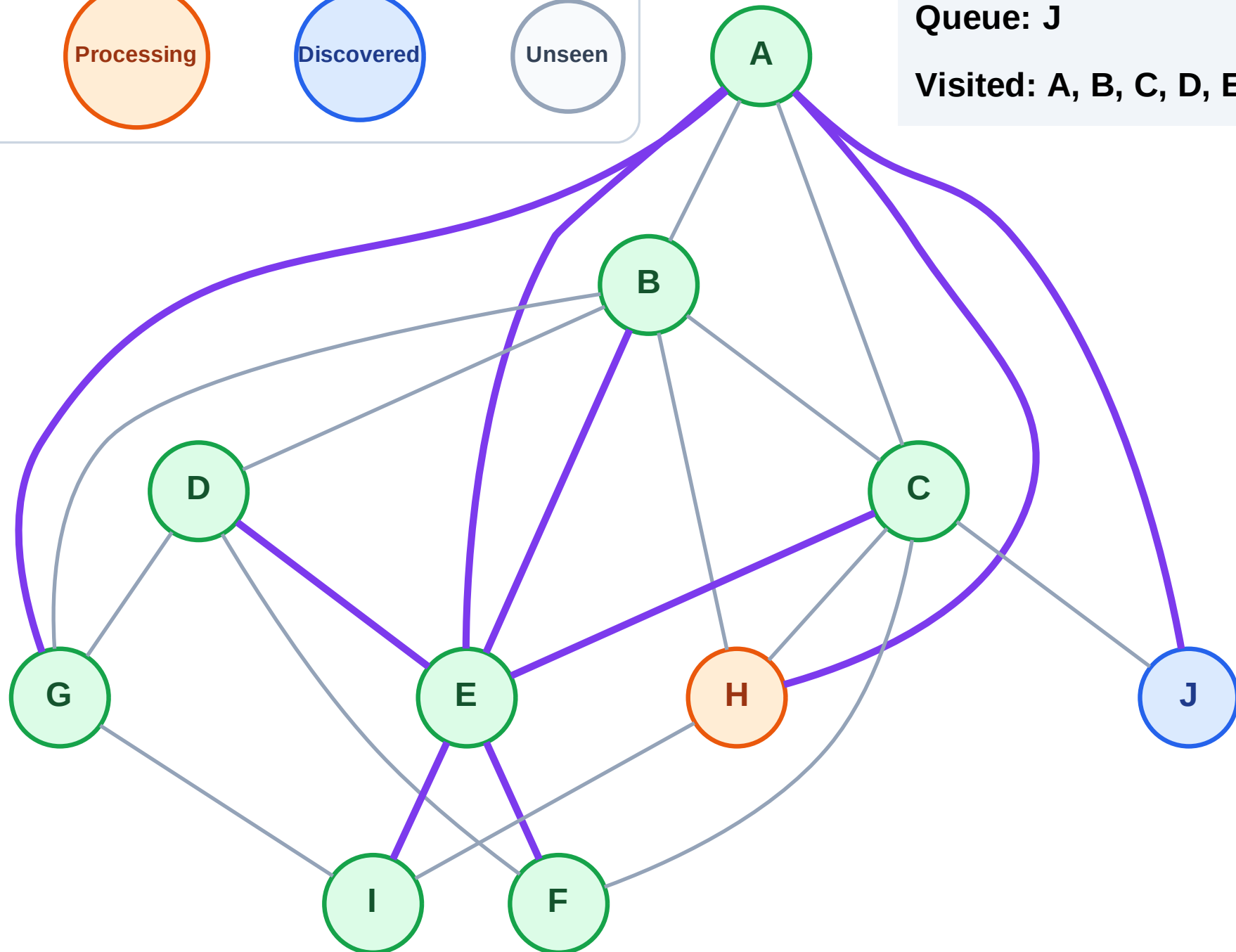
Step 18: Process node H

## Legend



Queue: J

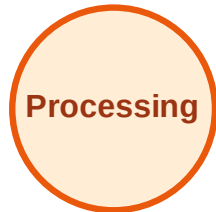
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

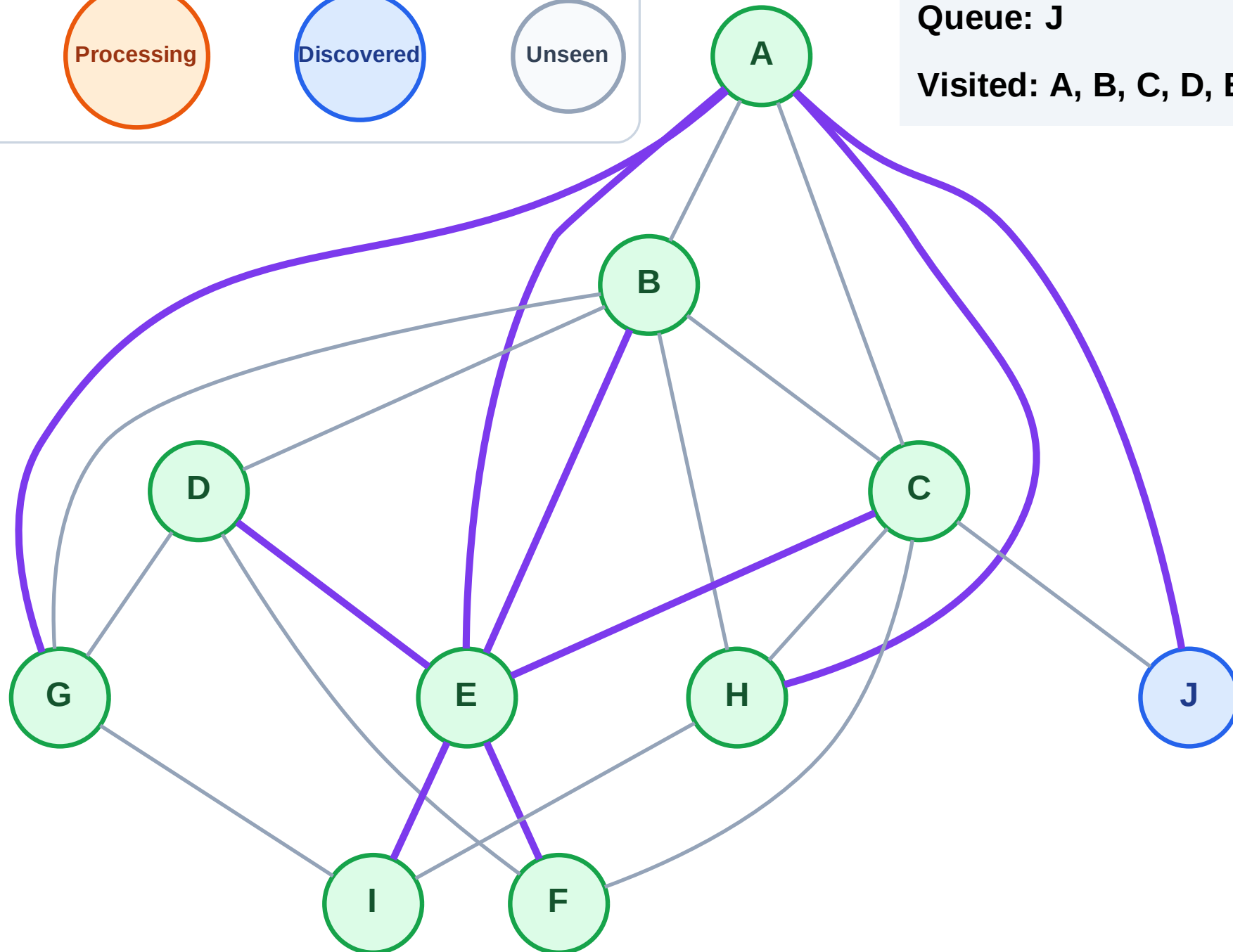
Step 19: No new neighbors from H

## Legend



Queue: J

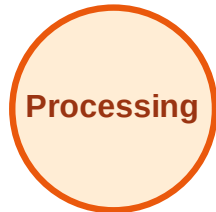
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

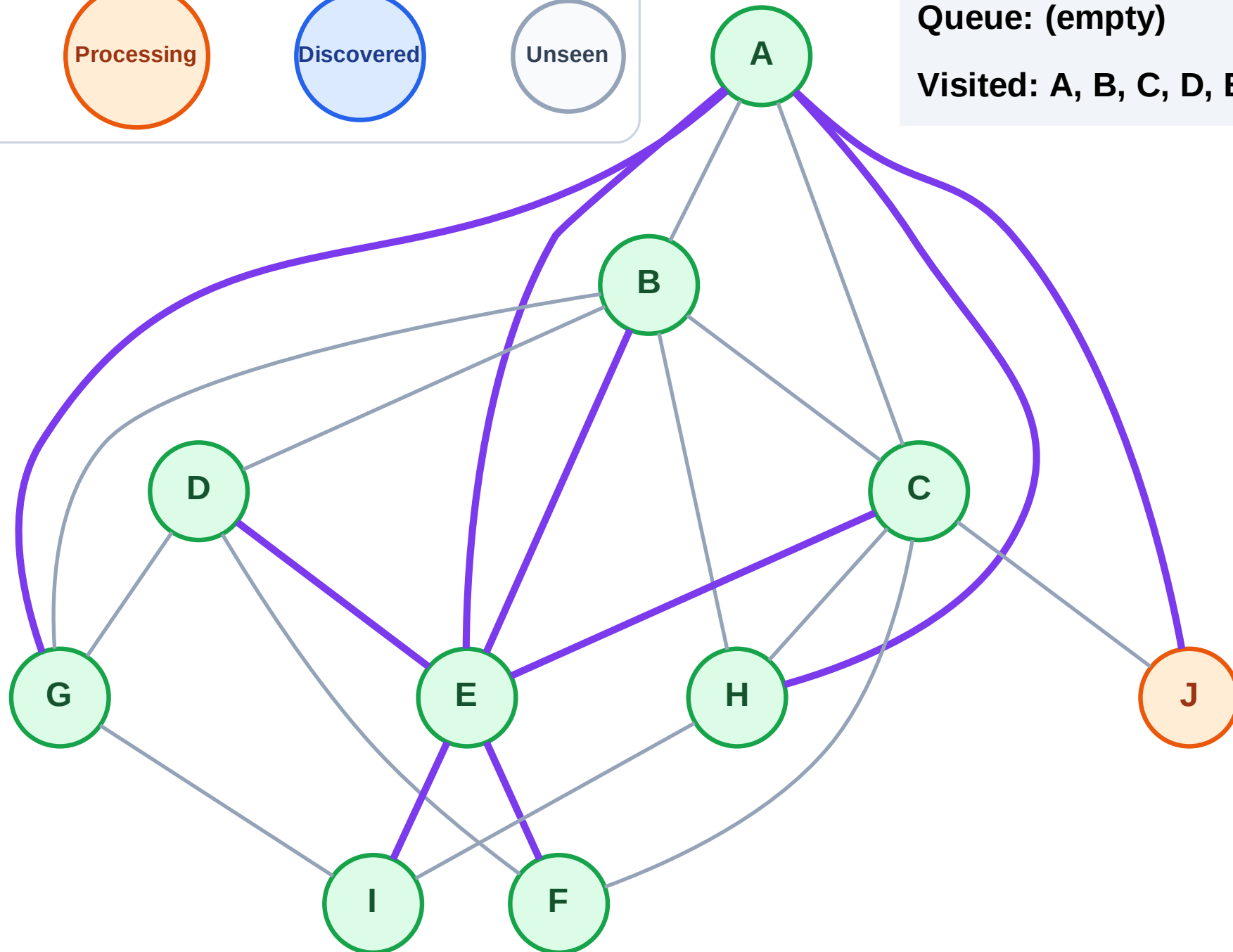
Step 20: Process node J

## Legend



Queue: (empty)

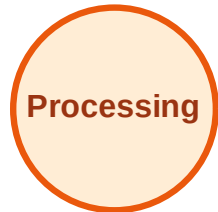
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

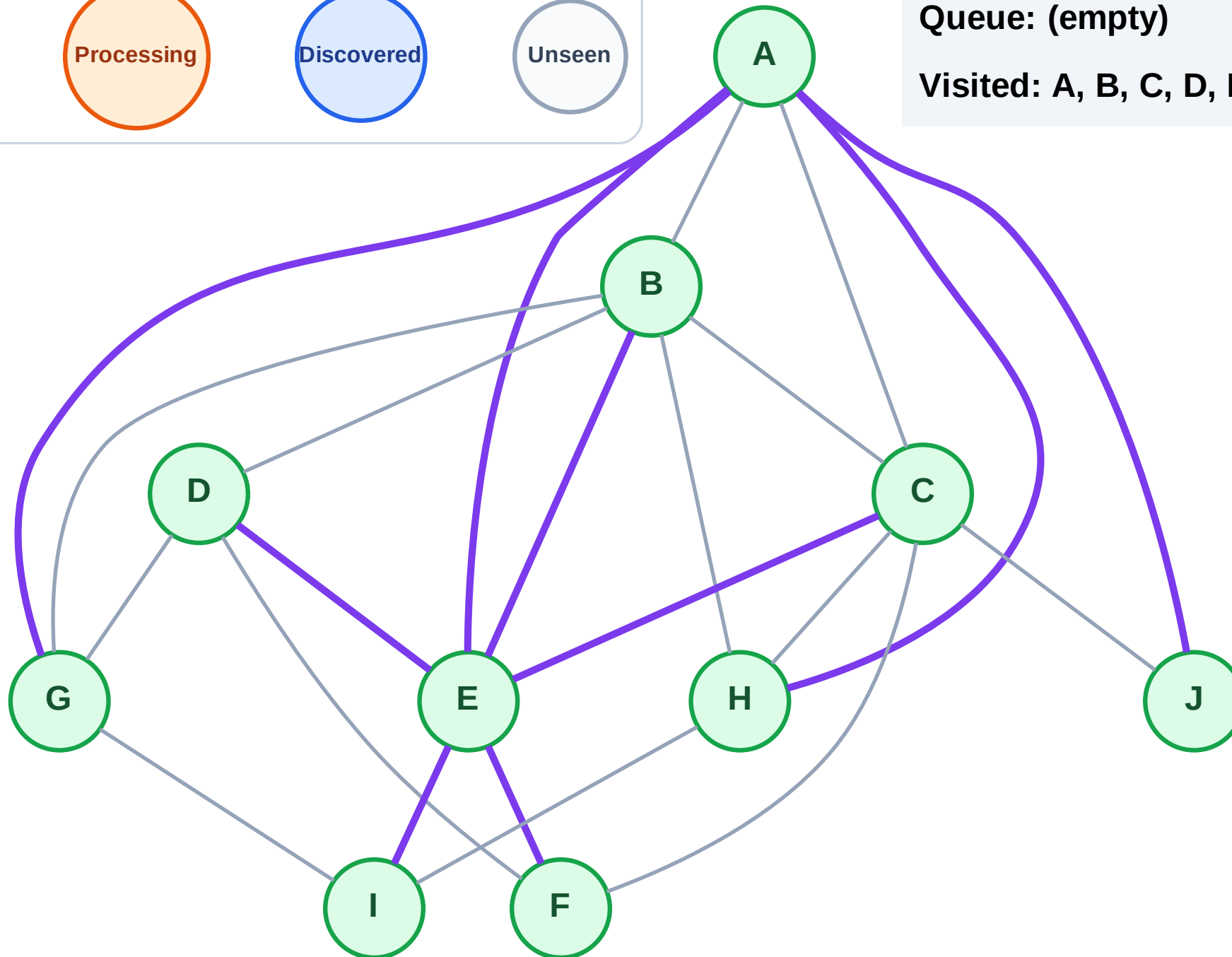
Step 21: No new neighbors from J

## Legend



Queue: (empty)

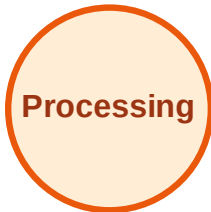
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

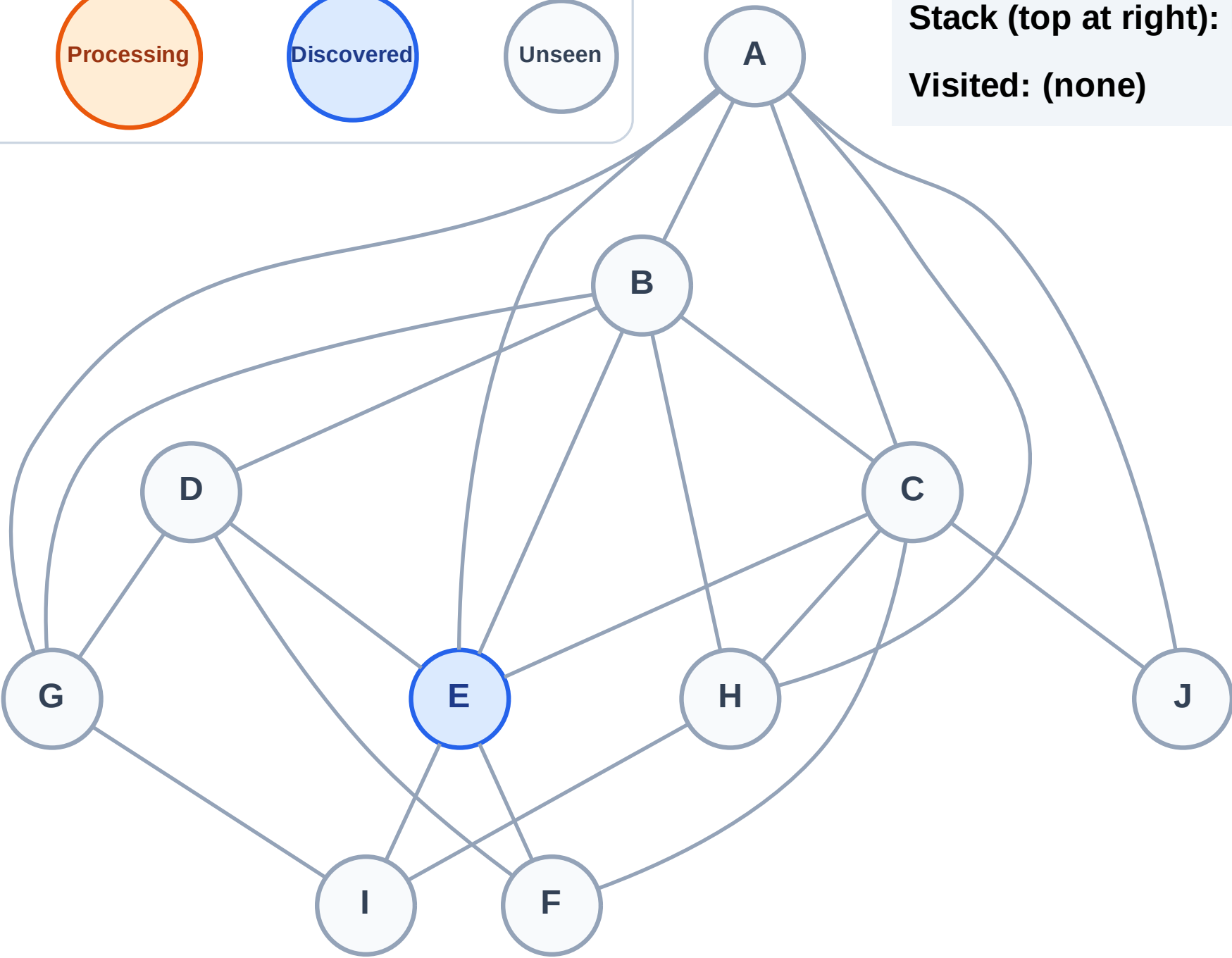
Step 1: Start at E

## Legend



Stack (top at right): E

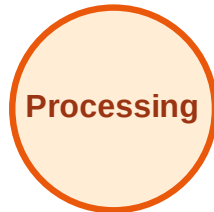
Visited: (none)



# DFS Traversal

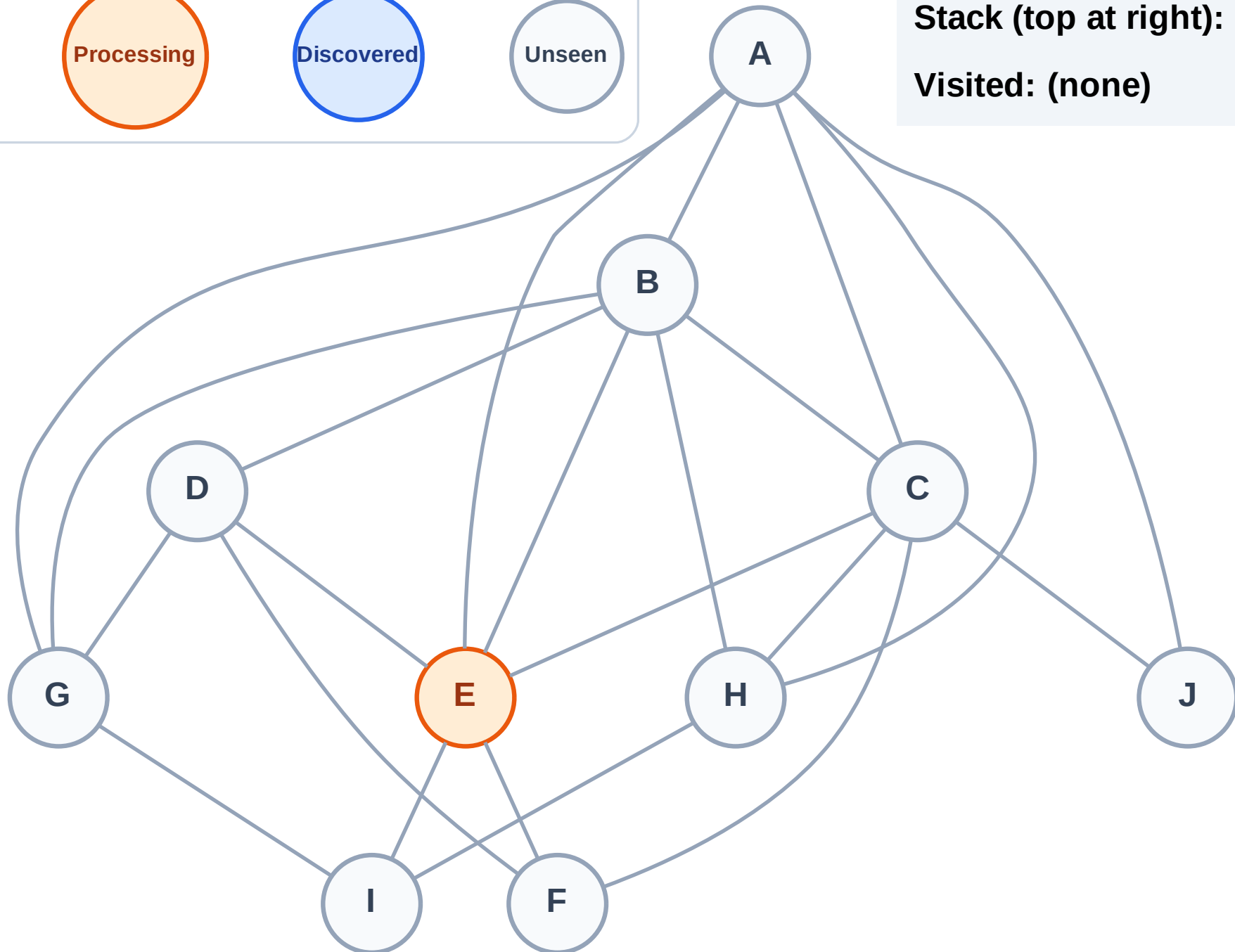
Step 2: Process node E

## Legend



Stack (top at right): (empty)

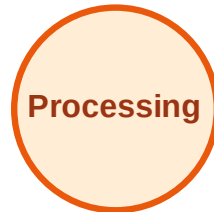
Visited: (none)



# DFS Traversal

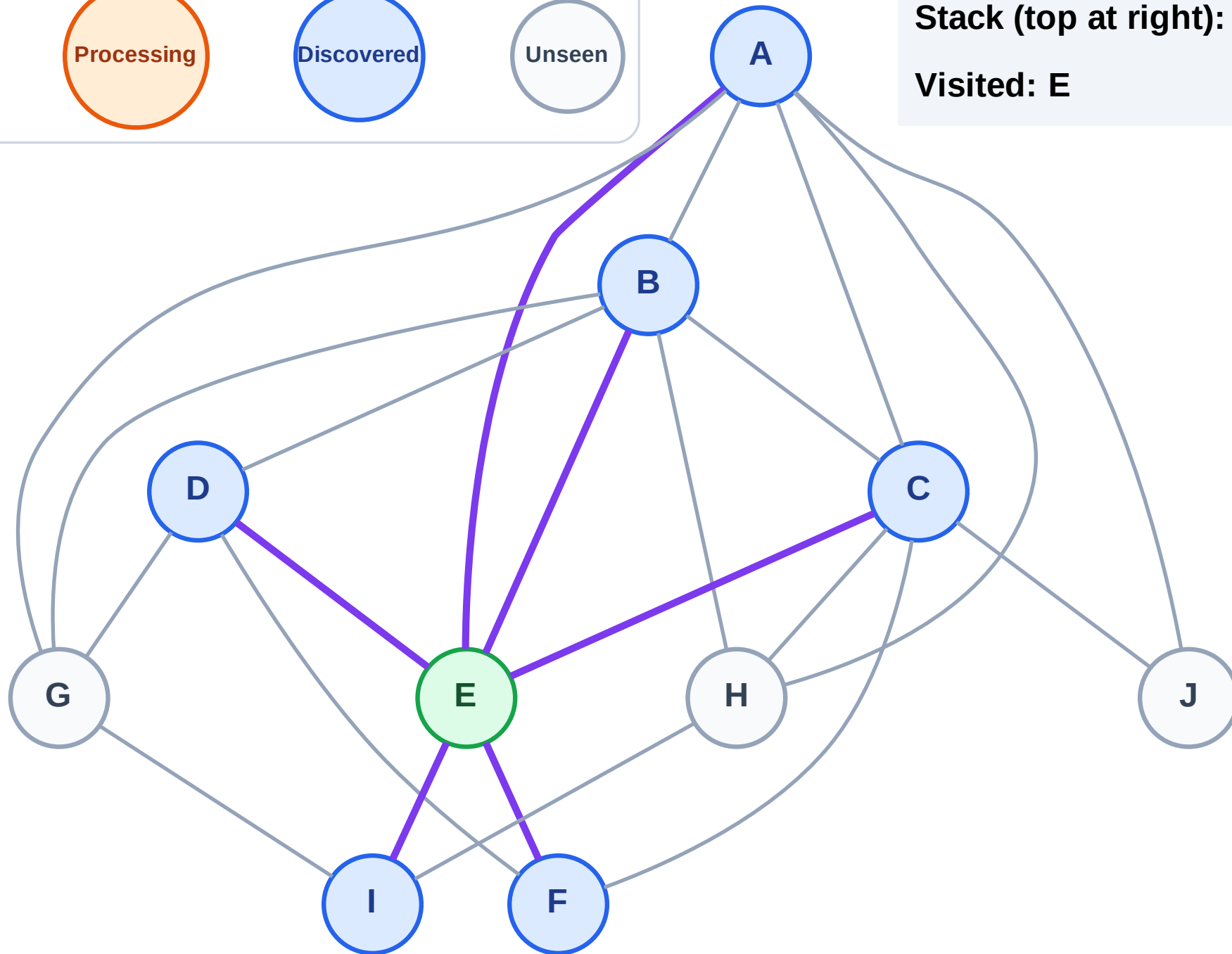
Step 3: Discover I, F, D, C, B, A from E

## Legend



Stack (top at right): I | F | D | C | B | A

Visited: E

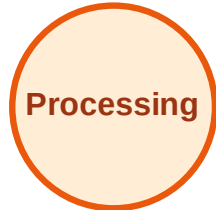




# DFS Traversal

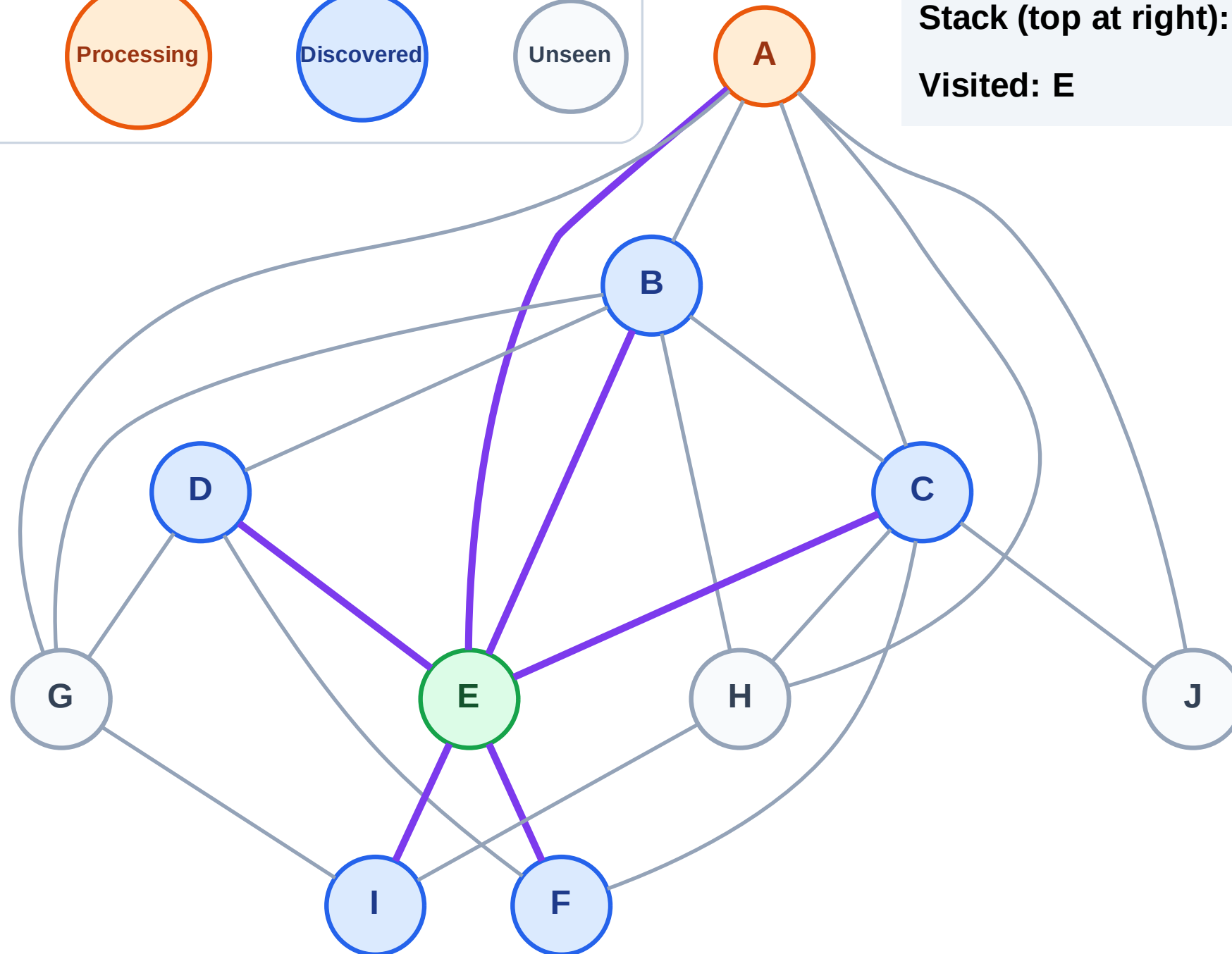
Step 4: Process node A

## Legend



Stack (top at right): I | F | D | C | B

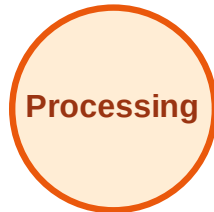
Visited: E



# DFS Traversal

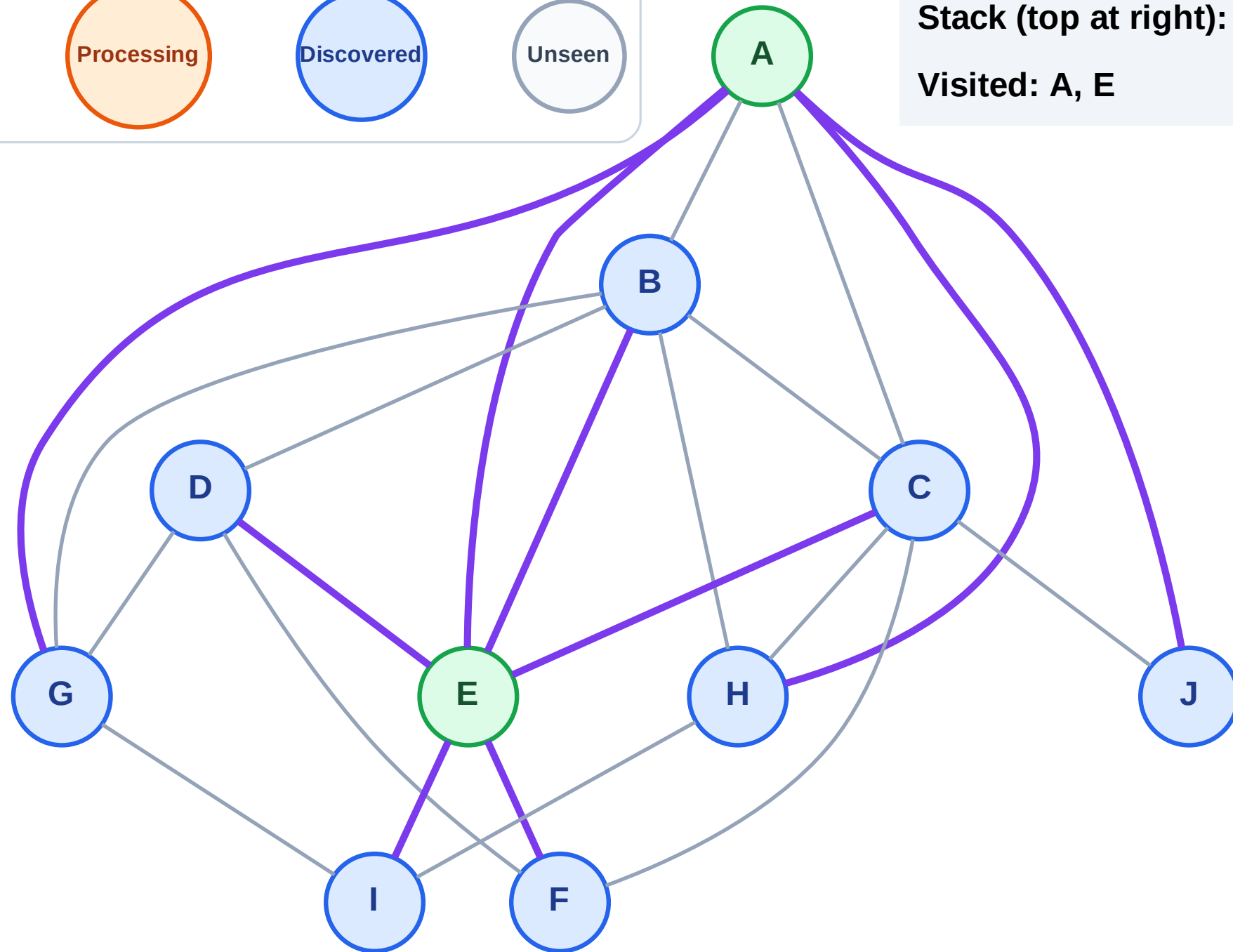
Step 5: Discover J, H, G from A

## Legend



Stack (top at right): I | F | D | C | B | J | H | G

Visited: A, E



# DFS Traversal

Step 6: Process node G

Legend

Complete

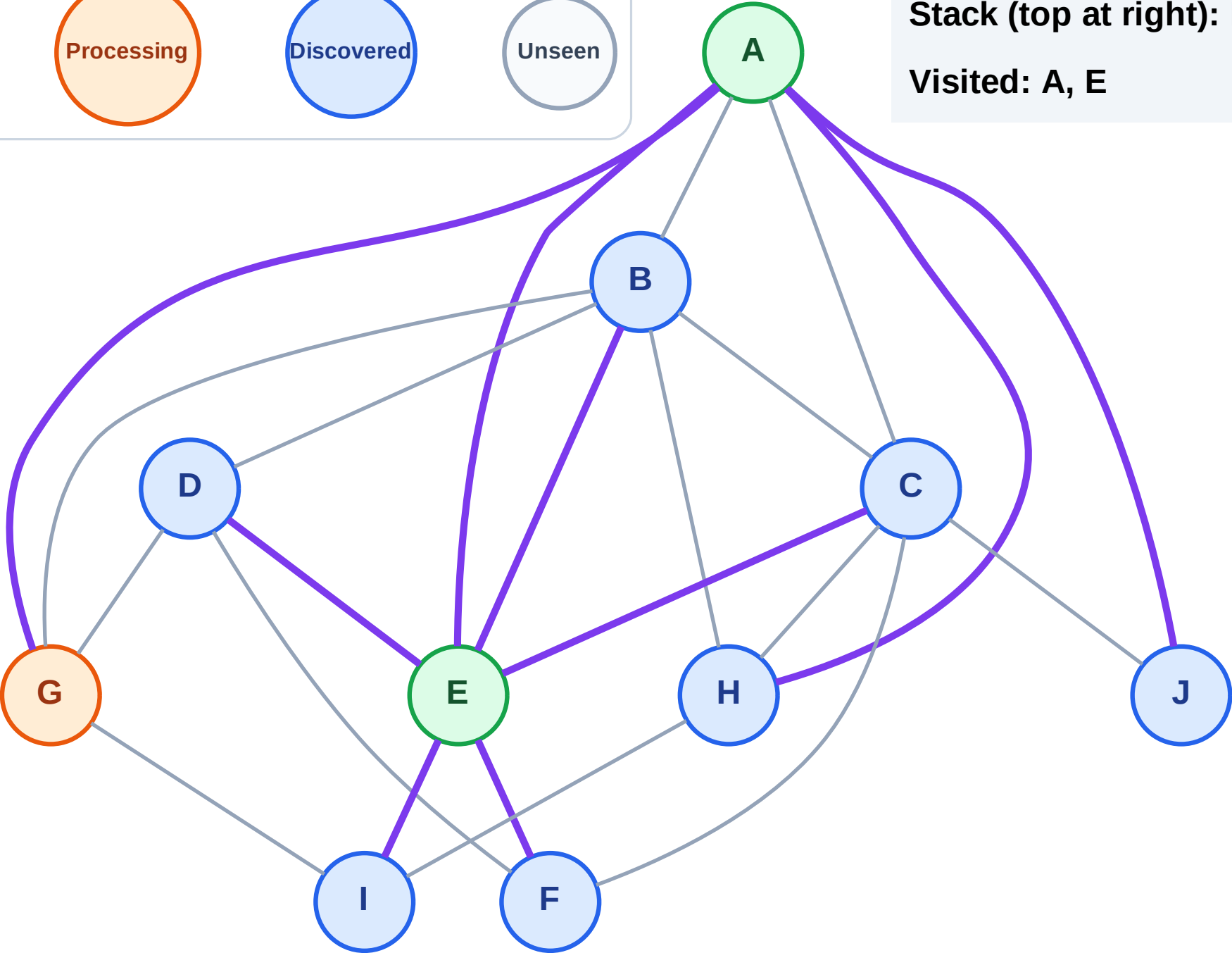
Processing

Discovered

Unseen

Stack (top at right): I | F | D | C | B | J | H

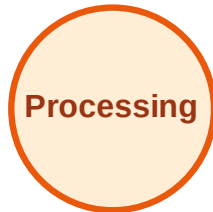
Visited: A, E



# DFS Traversal

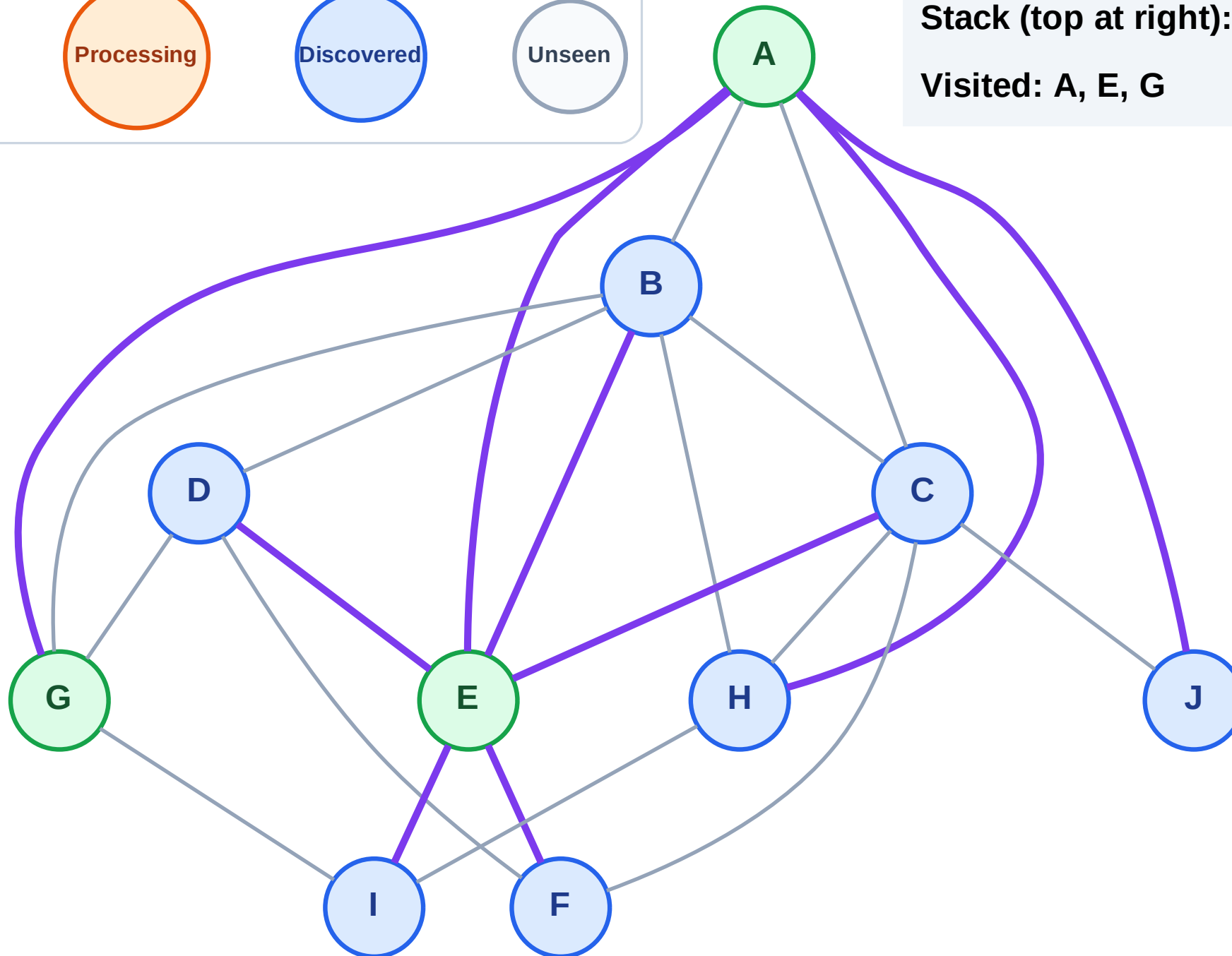
Step 7: No new neighbors from G

## Legend



Stack (top at right): I | F | D | C | B | J | H

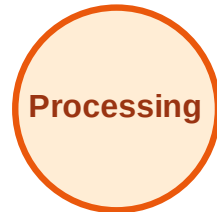
Visited: A, E, G



# DFS Traversal

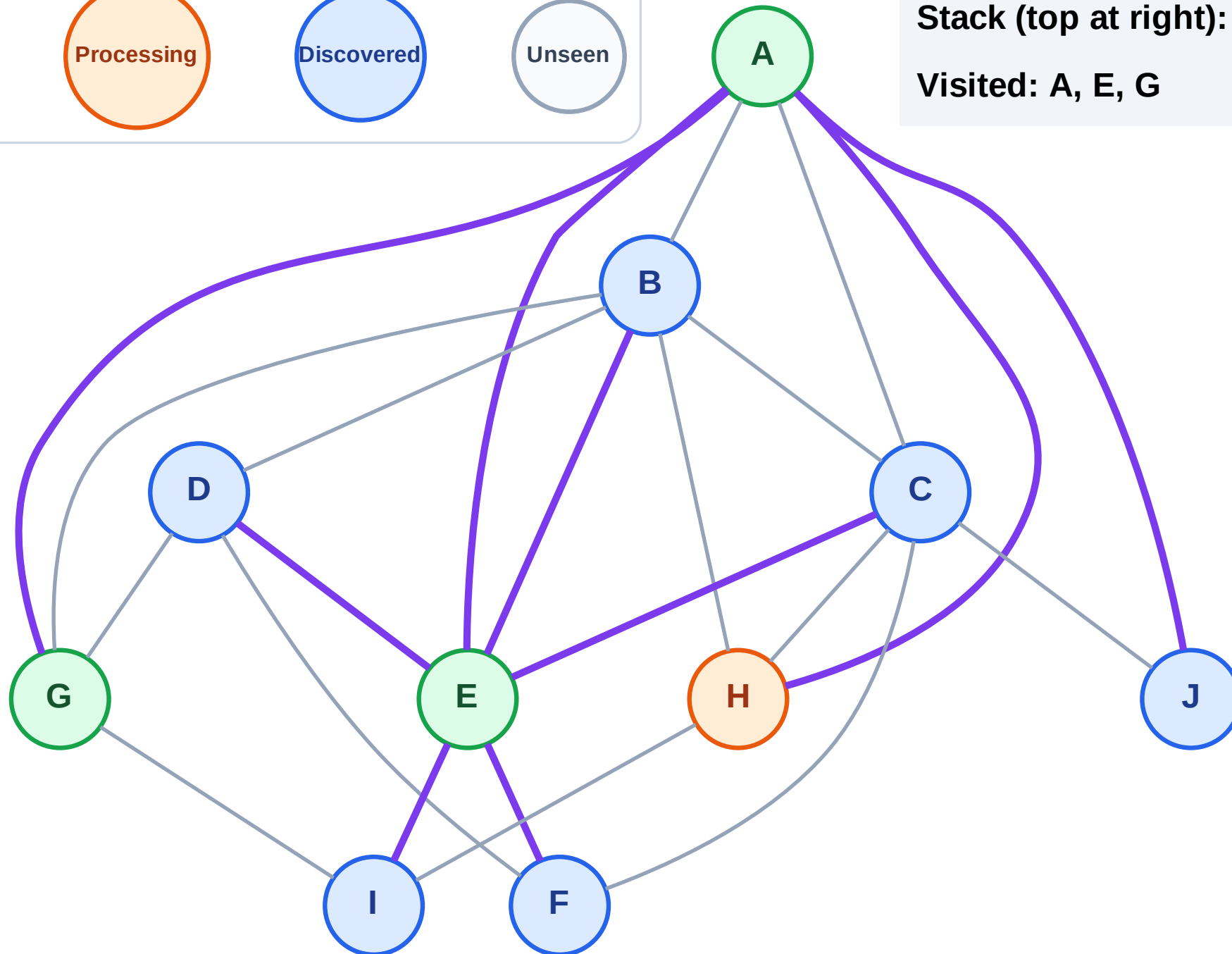
Step 8: Process node H

## Legend



Stack (top at right): I | F | D | C | B | J

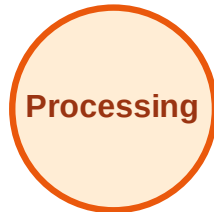
Visited: A, E, G



# DFS Traversal

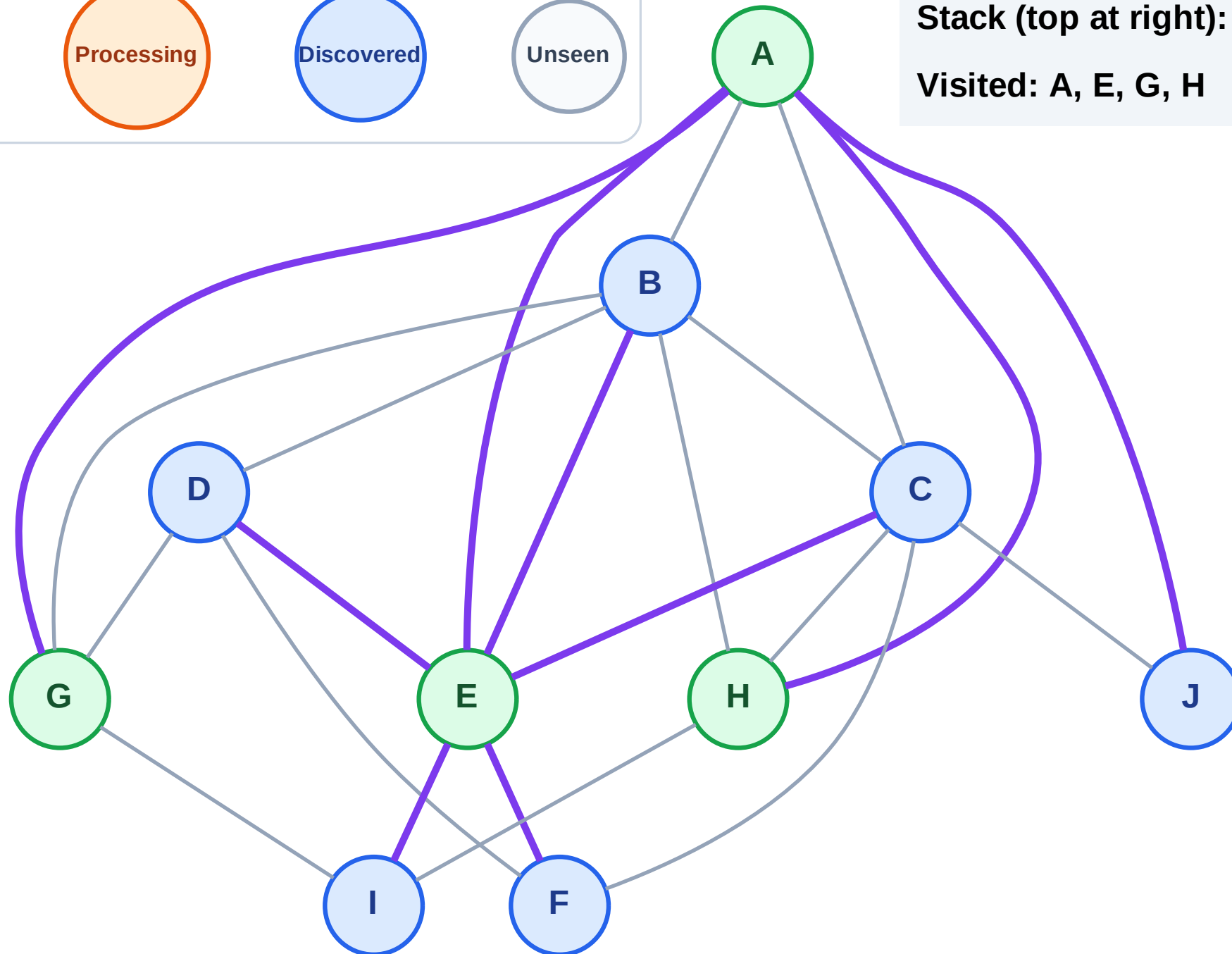
Step 9: No new neighbors from H

## Legend



Stack (top at right): I | F | D | C | B | J

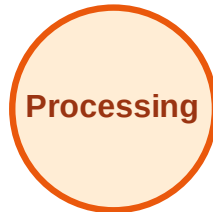
Visited: A, E, G, H



# DFS Traversal

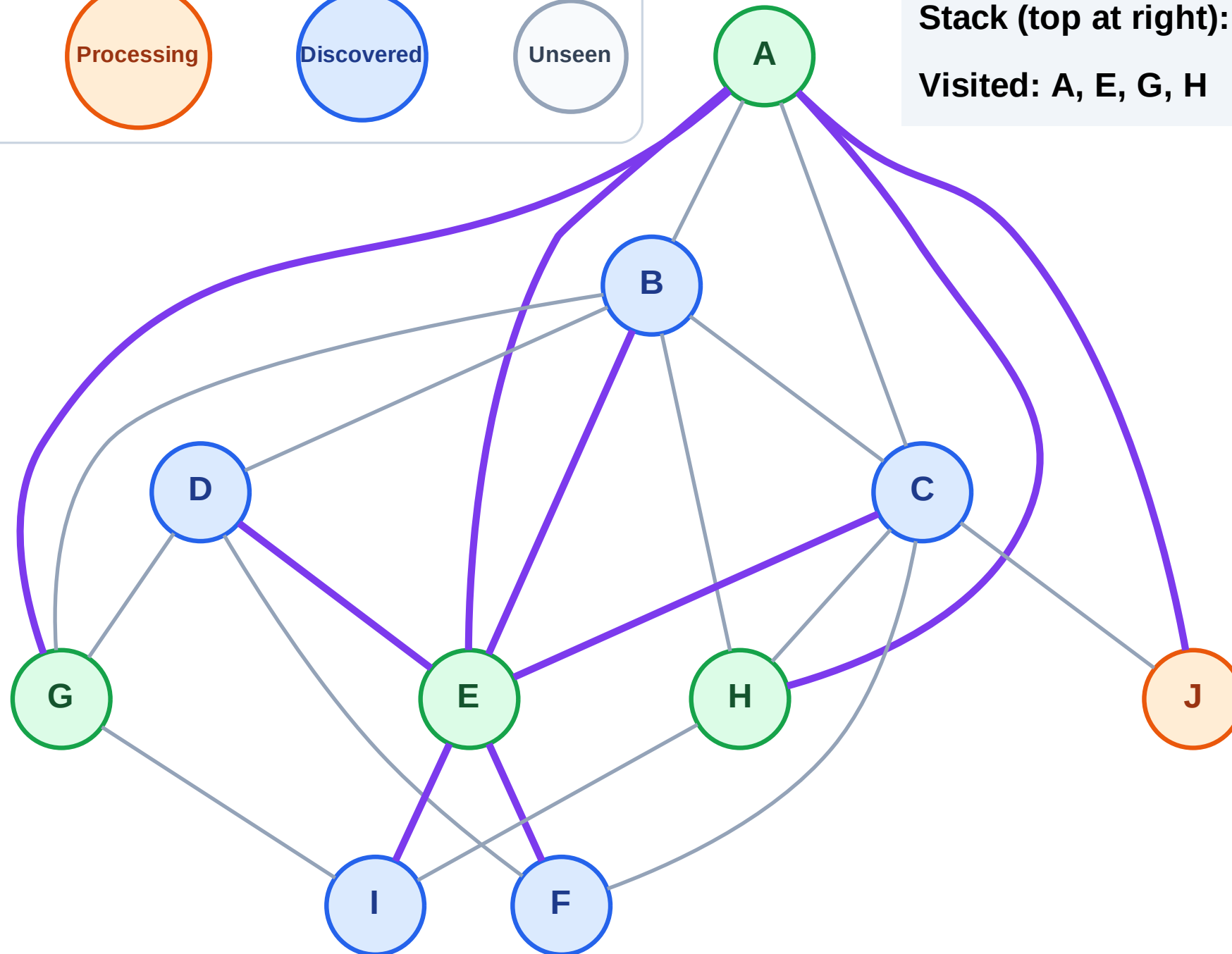
Step 10: Process node J

## Legend



Stack (top at right): I | F | D | C | B

Visited: A, E, G, H

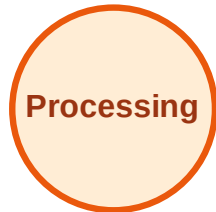




# DFS Traversal

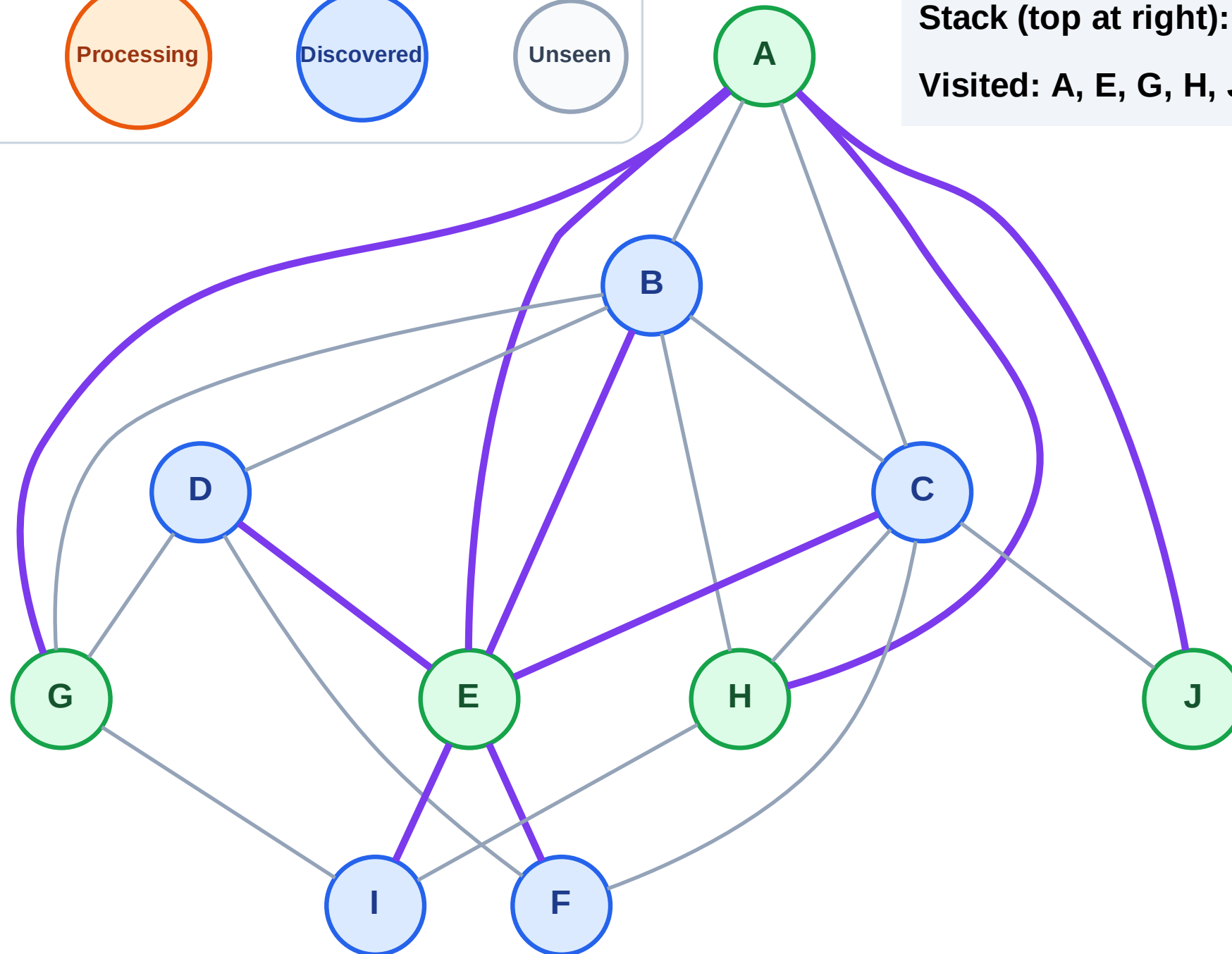
Step 11: No new neighbors from J

## Legend



Stack (top at right): I | F | D | C | B

Visited: A, E, G, H, J

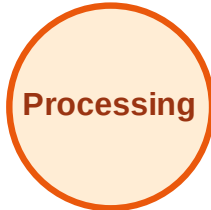




# DFS Traversal

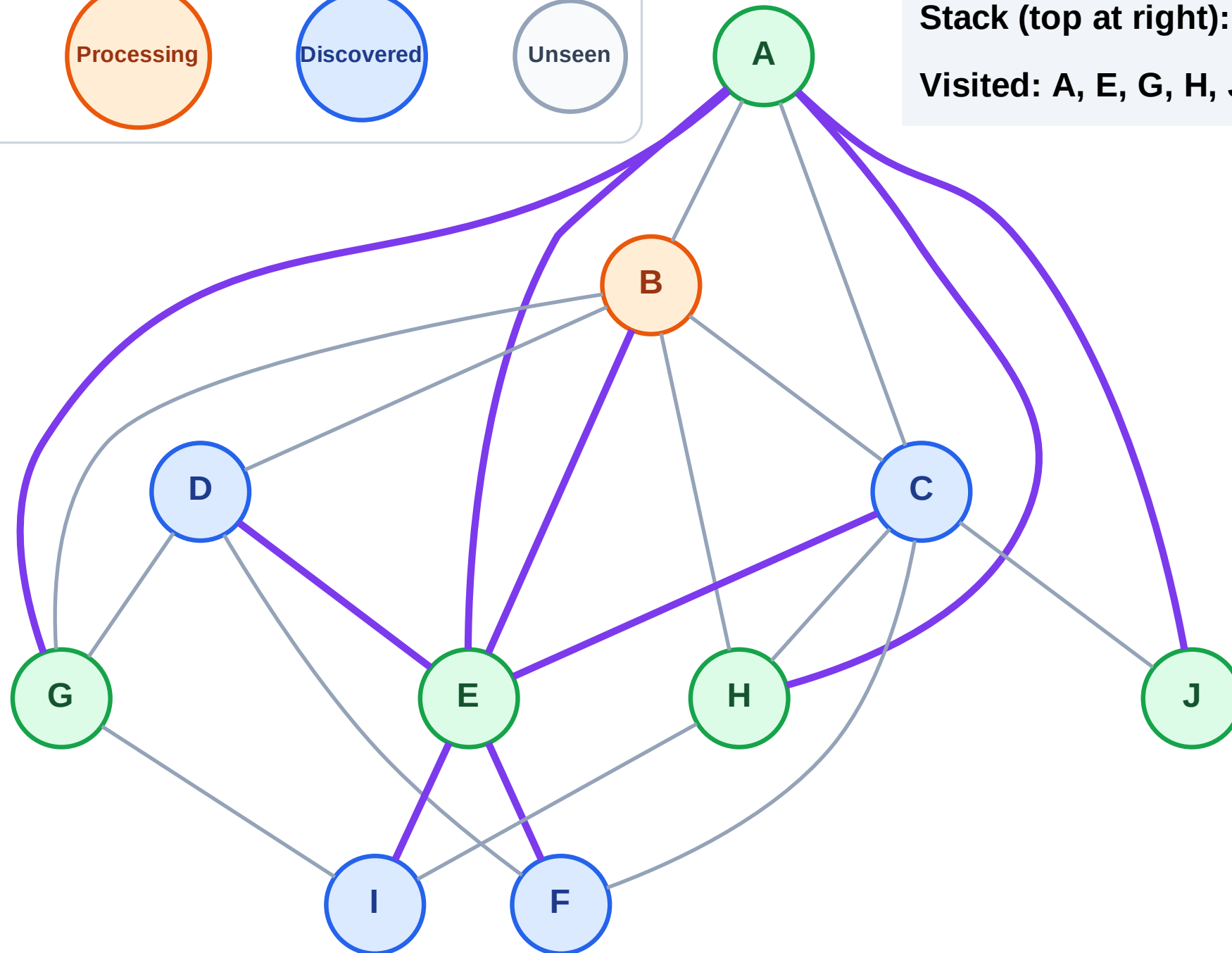
Step 12: Process node B

## Legend



Stack (top at right): I | F | D | C

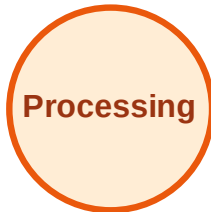
Visited: A, E, G, H, J



# DFS Traversal

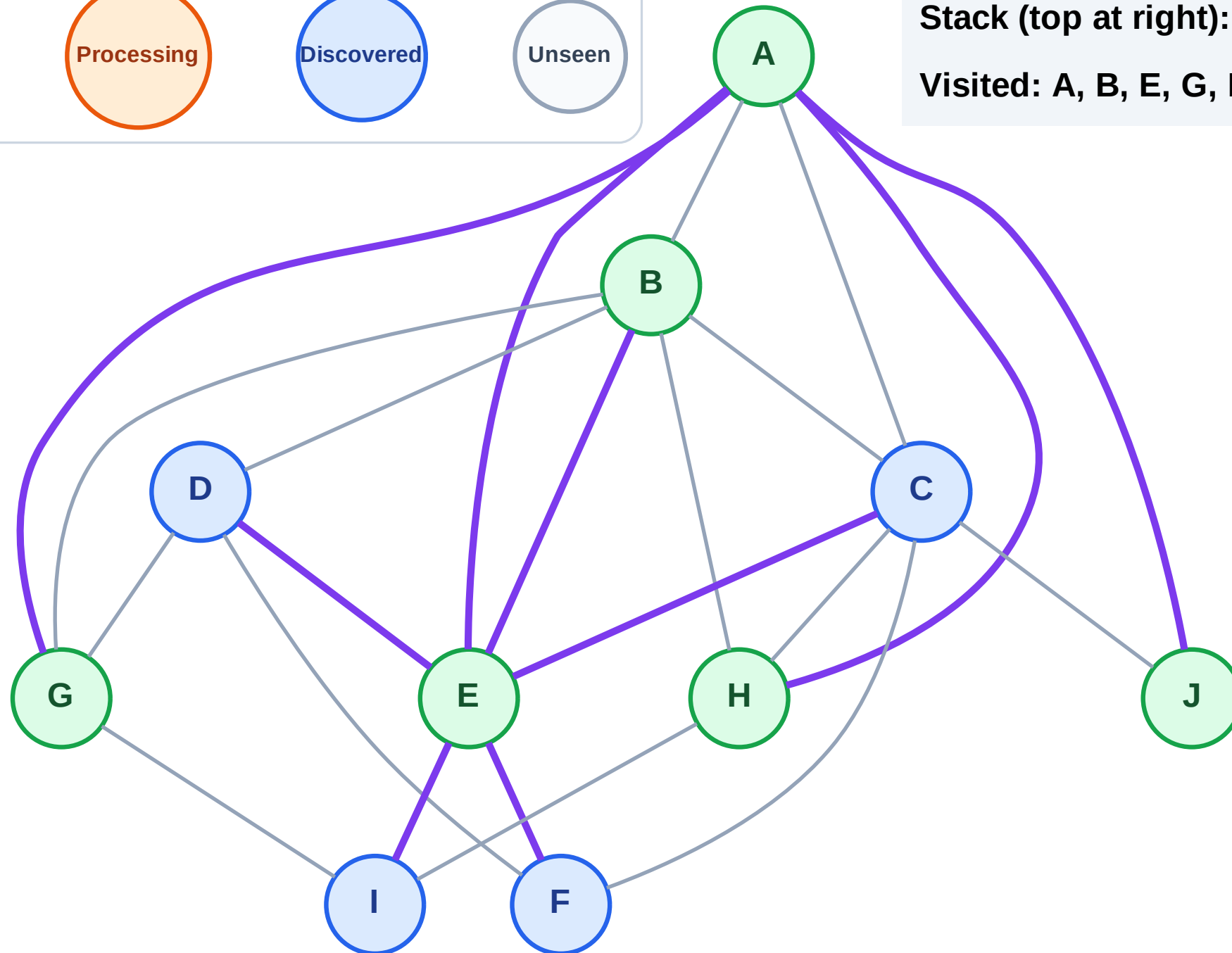
Step 13: No new neighbors from B

## Legend



Stack (top at right): I | F | D | C

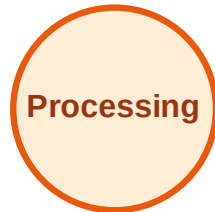
Visited: A, B, E, G, H, J



# DFS Traversal

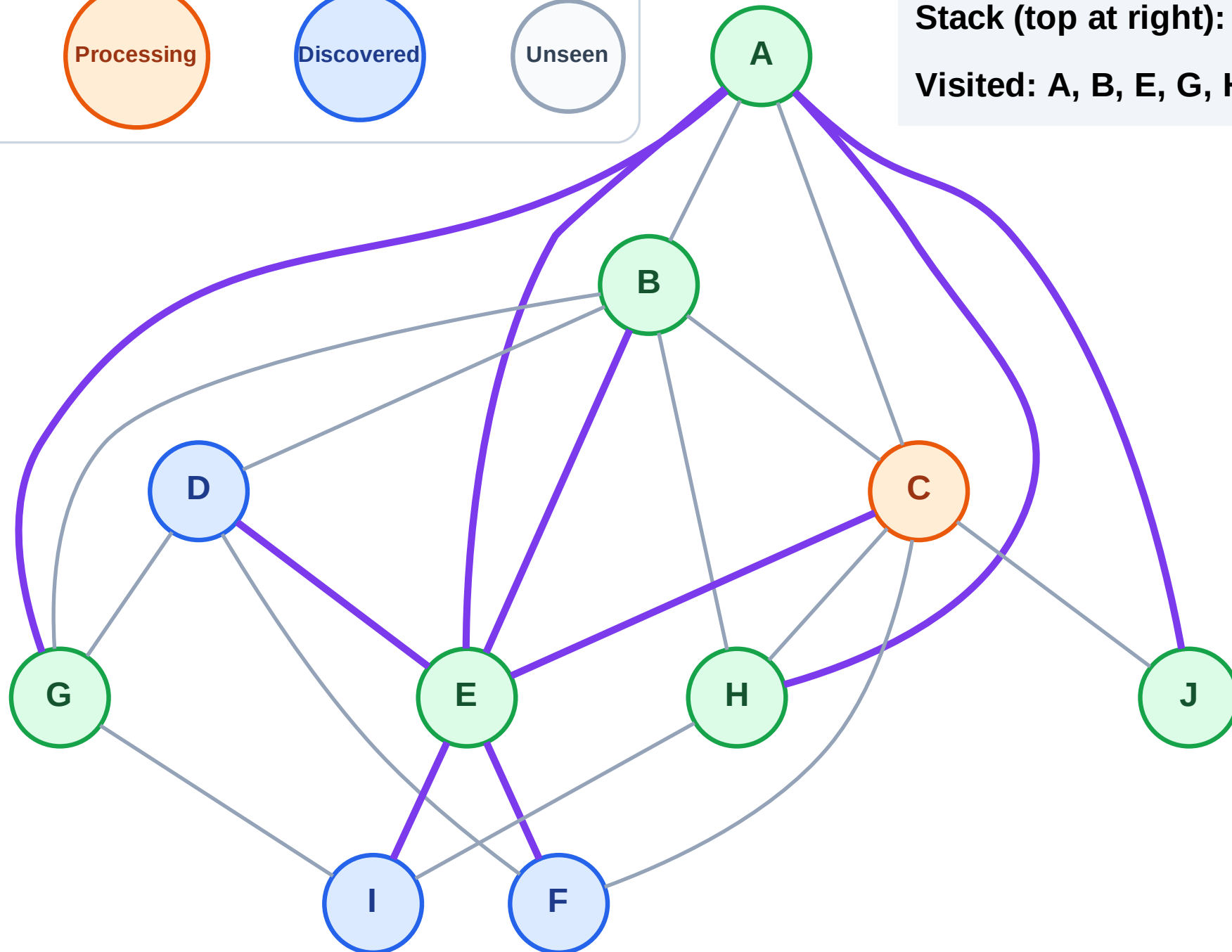
Step 14: Process node C

## Legend



Stack (top at right): I | F | D

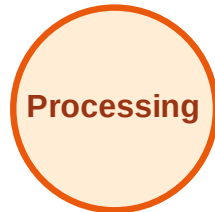
Visited: A, B, E, G, H, J



# DFS Traversal

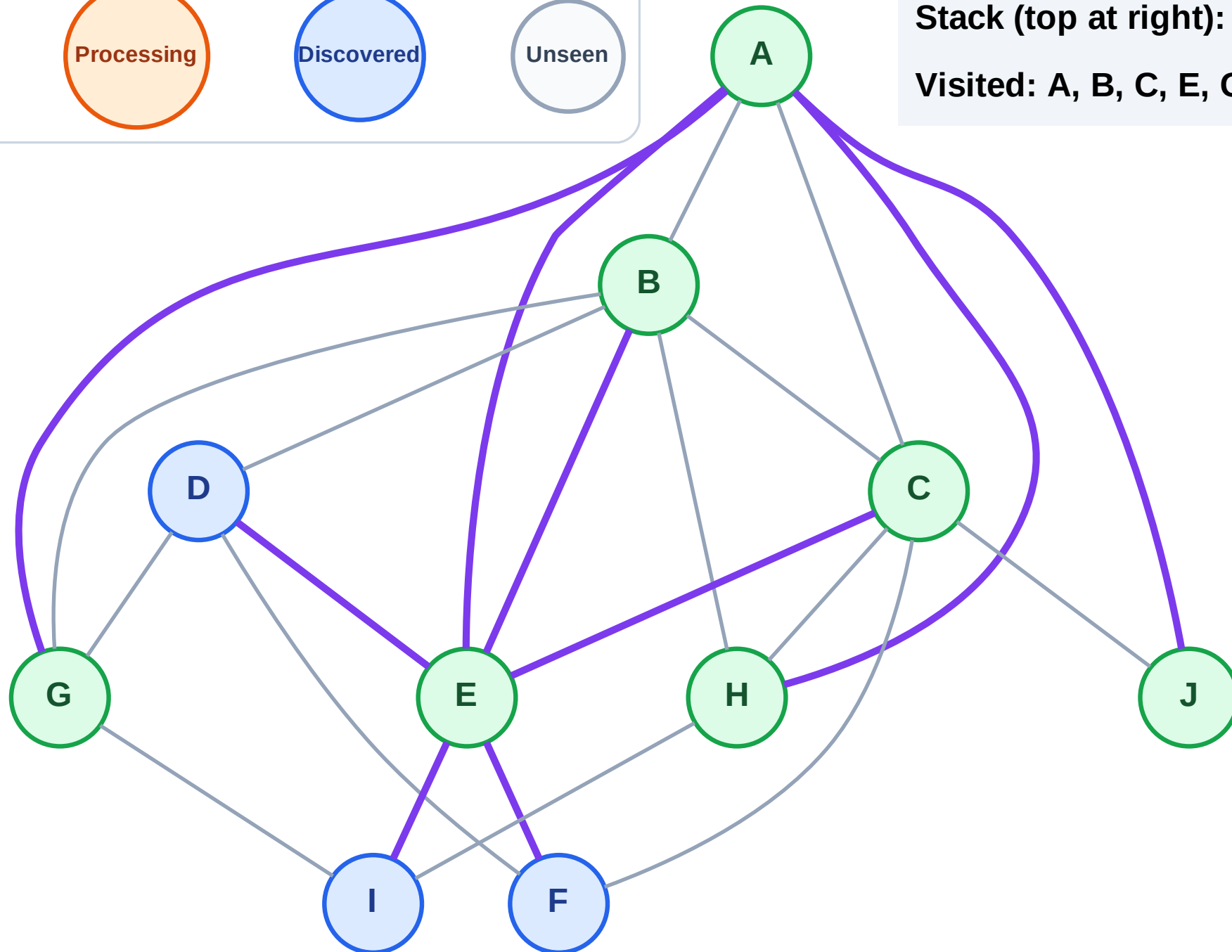
Step 15: No new neighbors from C

## Legend



Stack (top at right): I | F | D

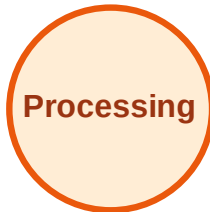
Visited: A, B, C, E, G, H, J



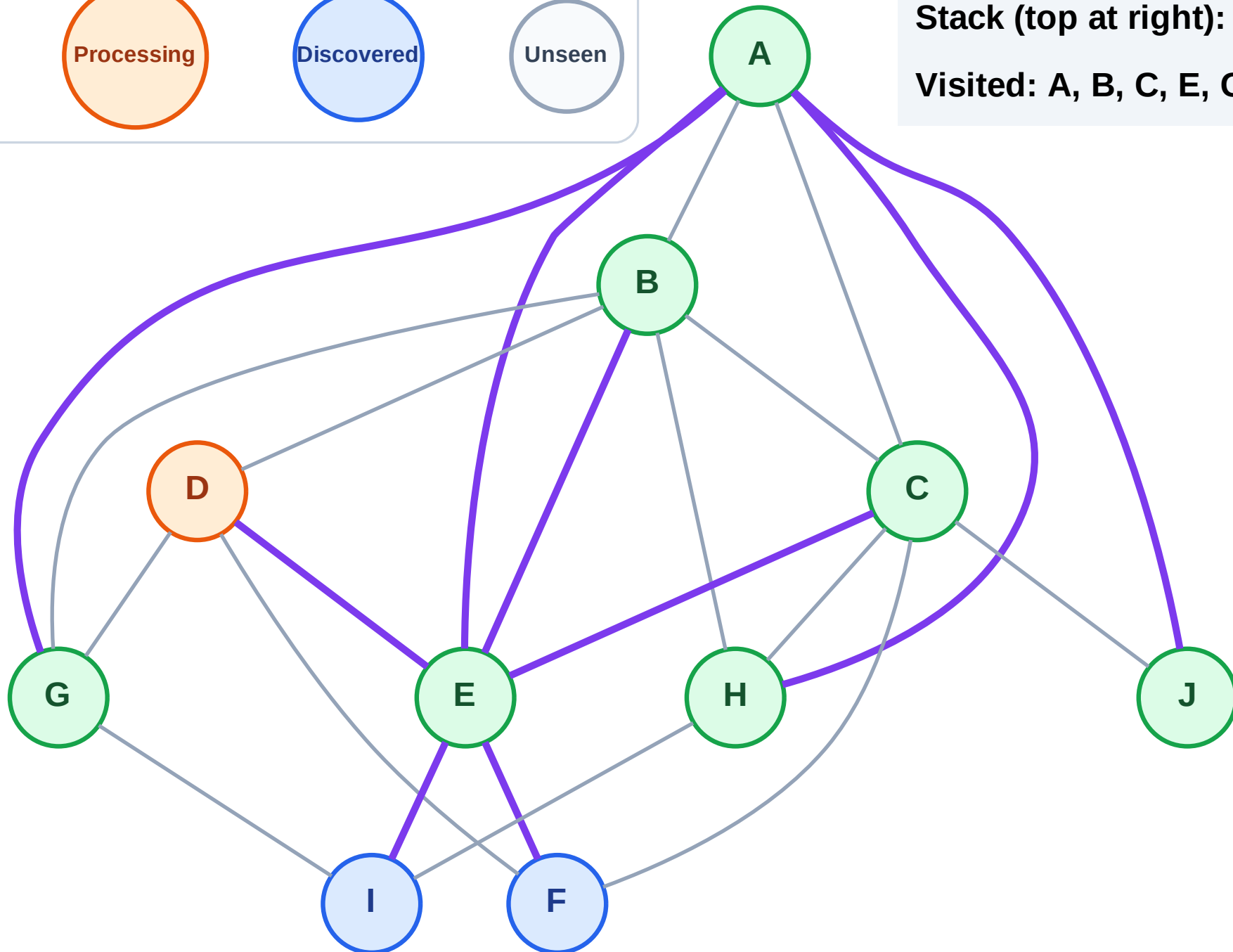
# DFS Traversal

Step 16: Process node D

## Legend



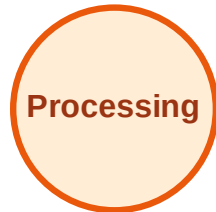
Stack (top at right): I | F  
Visited: A, B, C, E, G, H, J



# DFS Traversal

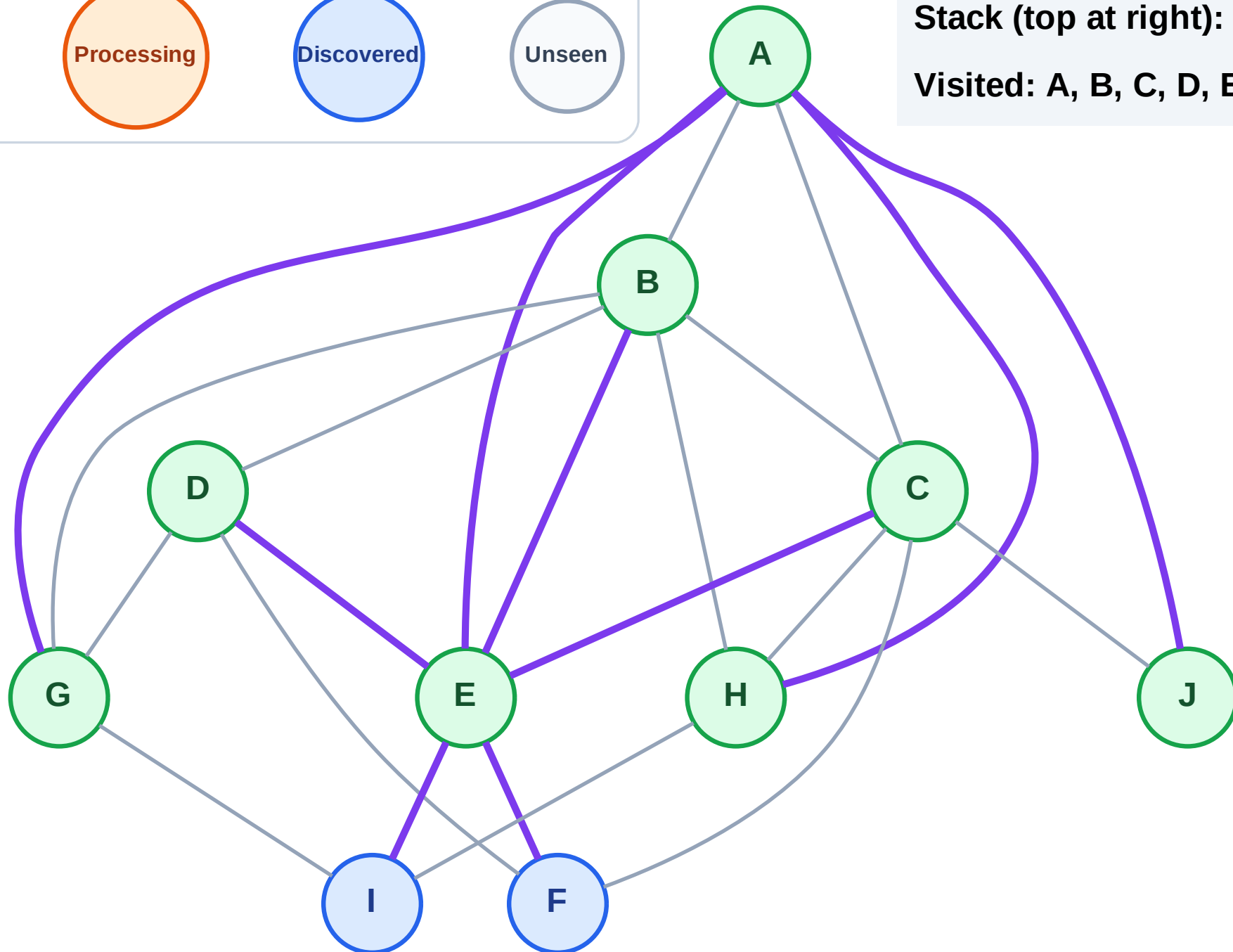
Step 17: No new neighbors from D

## Legend



Stack (top at right): I | F

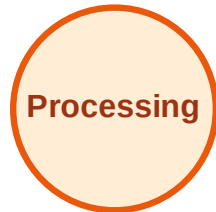
Visited: A, B, C, D, E, G, H, J



# DFS Traversal

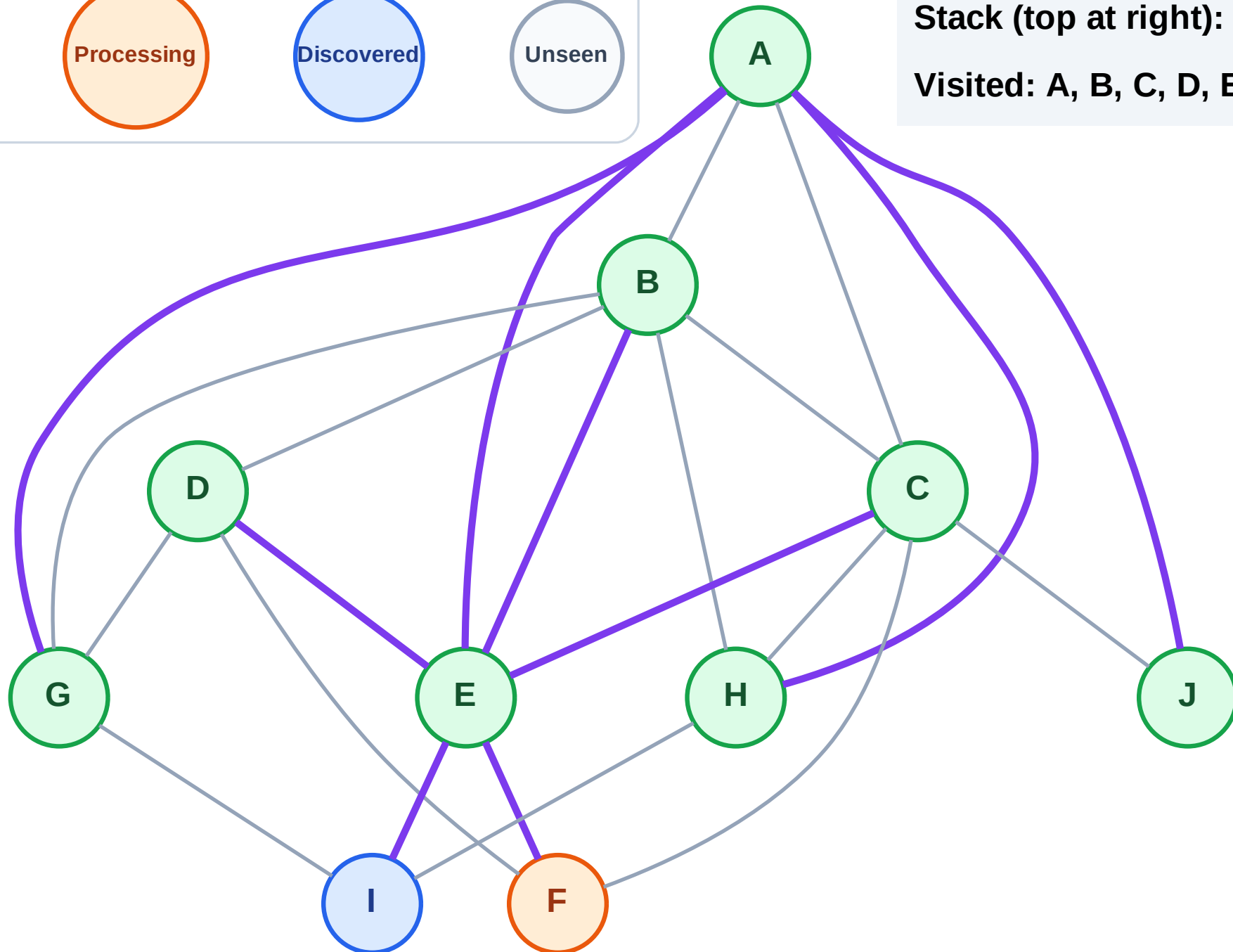
Step 18: Process node F

## Legend



Stack (top at right): I

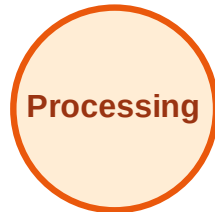
Visited: A, B, C, D, E, G, H, J



# DFS Traversal

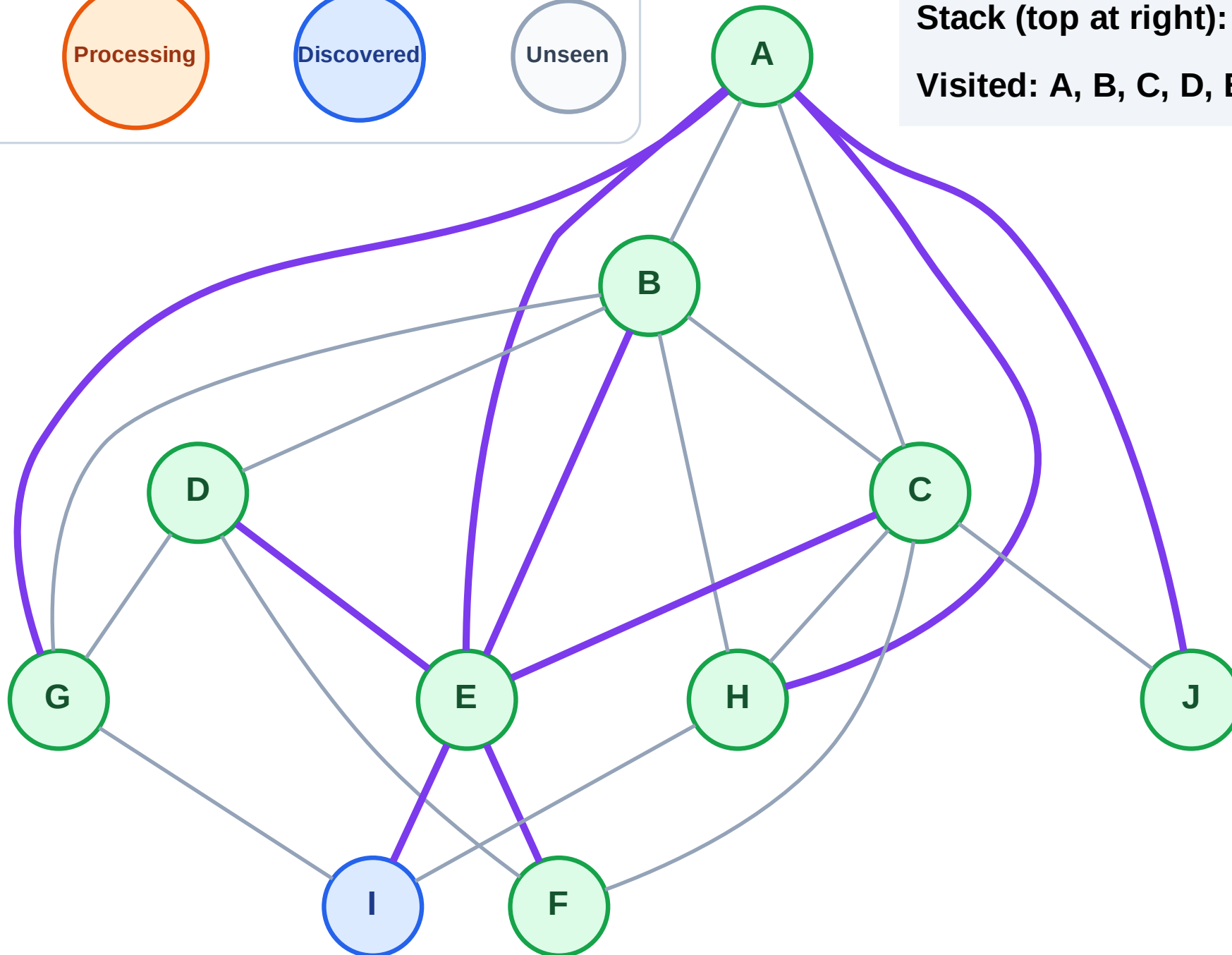
Step 19: No new neighbors from F

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

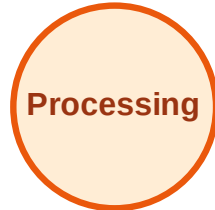




# DFS Traversal

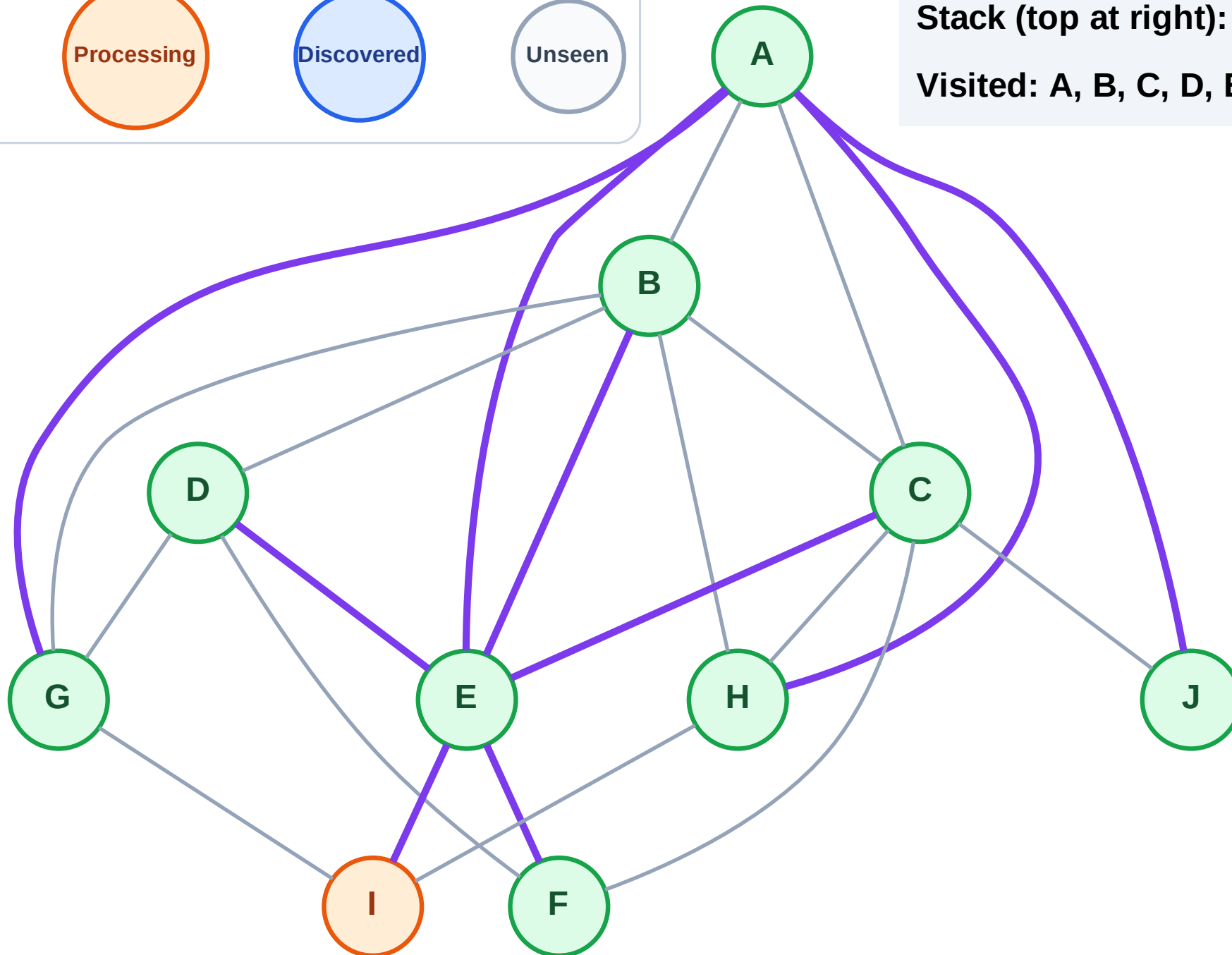
Step 20: Process node I

## Legend



Stack (top at right): (empty)

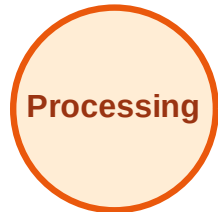
Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

